When K Matters: Model Selection in Finite Mixtures and the Magnitude of Policy Counterfactuals

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Abstract

The number of mixture components K in finite mixture models—typically treated as a technical detail—can be the dominant source of uncertainty in counterfactual policy analysis. I develop a three-pronged selection framework combining information criteria, counterfactual stability, and penalized estimation. In an application to bank branch access and self-employment, counterfactual predictions range from -0.6% ($K = 1$) to -10.8% ($K = 4$)—an 18-fold difference that dwarfs all other specification uncertainty by an order of magnitude. Monte Carlo simulations show this arises from a *cancellation condition*: when type-specific effects have opposite signs, the true K is recovered in fewer than 10% of replications. Rather than selecting a single K, I report bounds under model uncertainty and a Bayesian Dirichlet process posterior that integrates over K, yielding -8.5% (95% CI: $[-14.2\%, -3.1\%]$).

JEL Codes: C25, C38, C52, C11

Keywords: Finite mixture models, model selection, number of components, counterfactual analysis, unobserved heterogeneity, policy evaluation

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1 Introduction

Finite mixture models have become a workhorse tool in applied economics. Researchers use them to capture unobserved heterogeneity in treatment responses (Heckman and Singer, 1984), consumer preferences (Keane and Wolpin, 1997), dynamic discrete choices (Arcidiacono and Miller, 2011), labor market sorting (Bonhomme et al., 2019), and firm behavior (Fox et al., 2011). The practical challenge of selecting the number of mixture components K—too few types mask economically meaningful heterogeneity; too many invite overfitting and unstable estimates—is well understood in econometric theory (Kasahara and Shimotsu, 2009; Compiani and Kitamura, 2016; Chen et al., 2001). Yet in applied practice, K selection is typically relegated to a footnote: researchers report a BIC-preferred specification and conduct at most cursory sensitivity analysis.

This paper demonstrates that this practice can be dangerously misleading. In the application I study, assumptions about K swing counterfactual policy predictions by a factor of eighteen—dwarfing all other sources of specification uncertainty combined. The methodological insight is simple but underappreciated: when type-specific treatment effects have opposite signs, the aggregate counterfactual is a weighted average that depends critically on whether the analyst permits enough types to separate the underlying heterogeneity.

I develop and demonstrate a general framework for K selection in finite mixture counterfactual analysis, illustrated using the relationship between bank branch access and self-employment. A companion paper (Malkova, 2026) documents a robust reduced-form null for this application, providing a sharp benchmark for the structural analysis.

Monte Carlo simulations with two complementary designs reveal the severity of the problem. The "cancellation" design—where type-specific effects have opposite signs, producing a near-zero aggregate—generates within-replication counterfactual ranges of 15–17 pp across K, mirroring the empirical finding. Crucially, the true $K = 4$ is recovered in fewer than 10% of replications even at $N = 50{,}000$, because the cancellation mechanism makes types statistically near-indistinguishable. Under the "same-sign" control design, counterfactual magnitudes still vary with K but the qualitative conclusion is robust. These findings motivate the bounds and Bayesian model averaging approach developed below.

In the empirical application, a multinomial logit with $K = 4$ latent types identifies four economically distinct population segments, and the counterfactual jumps from -0.6% ($K = 1$) to -10.8% ($K = 4$). The 18-fold difference arises because the homogeneous model averages opposite-signed type-specific effects, producing a near-zero aggregate that obscures substantial within-type heterogeneity.

To discipline K selection, I develop a three-pronged approach combining three recent advances: (i) Hao and Kasahara (2025) panel BIC, which adjusts the penalty for repeated cross-sections of the same markets; (ii) Bonhomme, Lamadon, and Manresa (2022) counterfactual stability, which selects the smallest K at which the policy prediction stabilizes; and (iii) Budanova (2025) penalized MLE, approximated via overspecification and significance-based classification of active types. The framework converges on $K = 4$: while BIC marginally favors $K = 5$ (by 31 points, compared to a 1,460-point gap from $K = 3$ to $K = 4$), the overspecification test identifies the fifth type as redundant, confirming four active types. However, the counterfactual sensitivity analysis demonstrates that even well-motivated selection methods leave substantial residual model uncertainty.

Rather than reporting a single K-selected point estimate, I present three complementary summaries of the counterfactual:

1. *Bounds under model uncertainty*: $\Delta SE \in [-10.8\%, -0.6\%]$, valid for any $K \in \{1, \ldots, 5\}$.

2. *Bayesian posterior integrating over K*: mean -8.5%, 95% credible interval $[-14.2\%, -3.1\%]$, $P(\text{effect} < 0) = 0.99$.

3. *BIC-selected point estimate*: -10.8% under $K = 4$.

The discrepancy between the structural findings and the reduced-form null creates a genuine interpretive tension. The structural interpretation is that the reduced-form null reflects cancellation across types, not absence of effects. The skeptical interpretation is that the structural heterogeneity is an artifact of functional form—the mixture model discovers types that happen to separate demographic groups with different self-employment rates, with no causal credit channel. I do not resolve this tension; doing so requires exogenous variation in branch access that the data do not provide. Instead, I develop a framework for transparently reporting results under both views, demonstrating how bounds and Bayesian model averaging can accommodate irreducible model uncertainty.

1.1 Contributions

The primary contribution is demonstrating that K selection in finite mixture models can be the dominant source of uncertainty in counterfactual policy analysis. Table 10 (previewed here, presented in Section 8) decomposes the sources of specification uncertainty in the counterfactual: K selection generates a 19× range, while statistical sampling uncertainty (1.9–3.2×), discrete vs. continuous heterogeneity (1.1×), and reduced-form control specification (1.2×) are all an order of magnitude smaller. This finding implies that any applied paper using finite mixtures should report sensitivity to K alongside standard robustness checks.

The paper contributes to several literatures:

Finite mixture model selection. I extend the work of Kasahara and Shimotsu (2009), Compiani and Kitamura (2016), Hao and Kasahara (2025), Budanova (2025), and Bonhomme et al. (2022) by showing that even when multiple selection criteria agree on K, the counterfactual implications can differ sharply from reduced-form evidence. The three-pronged framework integrates complementary selection perspectives and makes residual model uncertainty transparent.

Structural vs. reduced-form methods. The tension between structural and reduced-form identification is a central methodological debate in economics (Heckman, 2010; Angrist and Pischke, 2010; Keane, 2010; Nevo and Whinston, 2010). I demonstrate a concrete setting where the two approaches yield qualitatively different policy conclusions and develop a framework for reporting results under both views.

Sensitivity analysis and model uncertainty. The paper connects to the growing literature on reporting model uncertainty in applied work (Leamer, 1983; Oster, 2019; Cinelli and Hazlett, 2020; Andrews et al., 2019), extending these tools to the mixture model context. The bounds-based approach I develop parallels partial identification methods (Manski, 2003) but operates over the model space rather than the parameter space.

Monte Carlo evidence on K selection. The Monte Carlo simulations contribute to the finite sample understanding of mixture model selection criteria. The two-design comparison identifies a simple sufficient condition—opposite-signed type effects—under which K selection dominates all other specification choices.

The paper proceeds as follows. Section 2 reviews the literature on K selection. Section 3 presents the model selection framework. Section 4 examines K recovery and counterfactual sensitivity via Monte Carlo simulation. Section 5 describes the empirical application. Section 6 presents the main results. Section 7 discusses the reduced-form/structural tension. Section 8 quantifies the sources of specification uncertainty. Section 9 validates the results using mixed logit, Bayesian DPM, and conformal inference. Section 10 concludes with recommendations for applied practice.

2 Related Literature on K Selection

2.1 Theoretical Foundations

Testing the number of components in mixture models is complicated by nonstandard asymptotics: under $H_0 : K = K_0$, the likelihood ratio statistic does not have a χ^2 distribution because parameters of the additional component lie on the boundary of the parameter space (Chen et al., 2001; Chen and Li, 2009). Kasahara and Shimotsu (2009) develop a modified likelihood ratio test for panel data that exploits the additional information from repeated observations, showing that repeated measurements restore standard asymptotics for the LR test. Kasahara and Shimotsu (2014) extend this to dynamic models with state dependence.

Information criteria offer a practical alternative. The standard BIC (Schwarz, 1978) penalizes model complexity by $p \cdot \ln(N)$, where p is the number of free parameters and N the sample size. Compiani and Kitamura (2016) provide a comprehensive treatment for economic applications, showing that BIC performs well in simulations but can be sensitive to sample size and parameter dimensionality. Hao and Kasahara (2025) develop a panel-specific BIC that replaces N with N_{panels} and adds a correction factor $c(T) = 1 + 1/T$ for the number of time periods:

$$\text{Panel BIC}(K) = -2 \ln \mathcal{L}(K) + p_K \cdot \ln(N_{panels}) \cdot c(T) \tag{1}$$

This adjustment addresses the tendency of standard BIC to over-select when applied to panel data, where the effective number of independent units is smaller than N.

2.2 Penalized and Bayesian Approaches

An alternative to information criteria is penalized likelihood, which shrinks redundant component shares toward zero. Budanova (2025) proposes penalized MLE for mixture models where the penalty term encourages sparsity in the mixing weights π_k. Under appropriate regularity conditions, the estimator consistently selects K while maintaining parametric convergence rates for the identified parameters.

Bayesian nonparametric methods avoid selecting K altogether by placing a prior over the space of mixing distributions. The Dirichlet Process Mixture (DPM) model (Ferguson, 1973; Antoniak, 1974) uses a Dirichlet process prior on the mixing distribution, allowing the effective number of components to be determined by the data. Malsiner-Walli et al. (2016) develop sparse finite mixture models that place a symmetric Dirichlet prior $\text{Dir}(e_0, \ldots, e_0)$ on mixing weights with $e_0 < 1$, encouraging sparsity so that redundant components empty out. This provides proper posterior uncertainty quantification over K without conditioning on a single selected value.

2.3 Counterfactual Stability

Bonhomme et al. (2022) propose a fundamentally different perspective: discrete types should be viewed as approximation devices for an underlying continuous distribution, not as literal population segments. Under this view, the "right" K is not a property of the data-generating process but a function of the research question. The appropriate criterion is counterfactual stability: the smallest K at which the counterfactual prediction of interest stops changing meaningfully. This approach directly targets the economic quantity of interest rather than statistical fit.

Bonhomme and Manresa (2015) develop a grouped fixed-effects estimator that assigns units to groups via k-means and estimates group-specific parameters jointly. Bonhomme et al. (2019) extend this to a distributional framework for matched employer-employee data. Both papers treat the number of groups as an approximation parameter chosen to balance bias and variance in the estimand of interest.

2.4 Gap in Applied Practice

Despite this rich theoretical literature, applied practice has lagged. In a survey of 40 papers using finite mixture or discrete heterogeneity models for counterfactual analysis published in leading economics journals (*Econometrica*, *AER*, *QJE*, *REStud*, *JPE*, *QE*, *JBES*) and related venues, I find that the modal approach to K selection is reporting BIC across 2–5 candidate values.[1] Only 4 of 34 applied papers (12%) fully report how the main counterfactual varies with K; 9 (26%) report partial sensitivity; the remaining 21 (62%) report only the selected-K point estimate.

[1]The survey covers articles identified via keyword search for "finite mixture," "latent class," "unobserved types," "number of components," "random coefficients," "discrete heterogeneity," and "grouped fixed effects," restricted to papers computing policy counterfactuals or treatment effects. The full classification is in Appendix Table A7.

Classifying each paper by whether the application plausibly exhibits opposite-signed type effects—the *cancellation condition* formalized in Section 7—reveals that cancellation is not exotic: 14 of 34 applied papers (41%) involve settings where cancellation is *likely* (e.g., health insurance with adverse selection and moral hazard, demand estimation with heterogeneous price responses), and 19 (56%) involve settings where it is *possible*. Only 1 paper (3%) involves a setting where cancellation is clearly unlikely. This suggests the problem documented here—that K selection can qualitatively change counterfactual conclusions—is relevant across a broad range of applied settings. This paper demonstrates why current practice is insufficient: the counterfactual in my application varies by a factor of 18 across plausible values of K.

3 A Three-Pronged Framework for Selecting K

I propose combining three complementary approaches to K selection, each addressing a different aspect of the problem.

Notation. Throughout, γ_k denotes the type-specific treatment effect for type k in the finite mixture model ($k = 1, \ldots, K$). In the empirical application, γ_k captures the effect of branch-based credit access on self-employment utility. In the mixed logit specification (Section 9), $\gamma_i \sim F(\bar{\gamma}, \sigma_\gamma^2)$ denotes the continuously distributed individual-level coefficient.

3.1 Prong 1: Information Criteria (Statistical Fit)

BIC selects the K that best balances goodness of fit against model complexity. When panel or repeated cross-section structure is available, the Hao and Kasahara (2025) panel BIC (1) provides a more appropriate penalty. I implement both standard and panel BIC and check agreement. Agreement strengthens confidence; disagreement signals that the penalty calibration is consequential and should be investigated.

3.2 Prong 2: Counterfactual Stability (Economic Relevance)

Following Bonhomme et al. (2022), I compute the counterfactual of interest for each K and identify the smallest value at which the prediction stabilizes. Formally, define the counterfactual as a function of K:

$$\Delta(K) = \frac{1}{N} \sum_{i=1}^{N} \left[\sum_{k=1}^{K} \hat{\pi}_k \cdot P(d_i = S | X_i, Z'; \hat{\theta}_k) - \sum_{k=1}^{K} \hat{\pi}_k \cdot P(d_i = S | X_i, Z; \hat{\theta}_k) \right] \tag{2}$$

where Z' denotes counterfactual conditions and Z denotes observed conditions. The stability criterion selects $K^* = \min\{K : |\Delta(K+1) - \Delta(K)| < \tau\}$ for a practical threshold τ. Bonhomme et al. (2022) suggest τ on the order of 1–2 percentage points of the counterfactual magnitude, though the specific threshold should be calibrated to the application.

3.3 Prong 3: Penalized MLE / Overspecification (Type Significance)

Following Budanova (2025), I start with an overspecified model (K larger than needed) and classify types as "active" or "redundant" based on the statistical significance of type-specific parameters. Types whose key parameters are indistinguishable from zero or from other types are candidates for merging. This approximates the penalized MLE where redundant type shares shrink to zero under the penalty term.

3.4 Integration

The three prongs address complementary questions:

Table 1: Three-Pronged Model Selection: Questions Addressed

Prong	Question	Criterion
1. BIC	Which K best fits the data?	Information criterion
2. Stability	Which K gives stable counterfactuals?	Policy prediction
3. OSCE	How many types are "real"?	Parameter significance

When all three agree, confidence in K selection is high. When they disagree, the pattern of disagreement is informative: BIC may favor more types than stability (overfitting on distributional features irrelevant to the counterfactual), or stability may require fewer types than OSCE (some statistically distinct types may have similar counterfactual implications). Disagreement should be reported transparently, and bounds across plausible K values should supplement the point estimate.

Figure 1 provides a visual summary of the three-pronged framework, showing how each criterion evaluates candidate values of K.

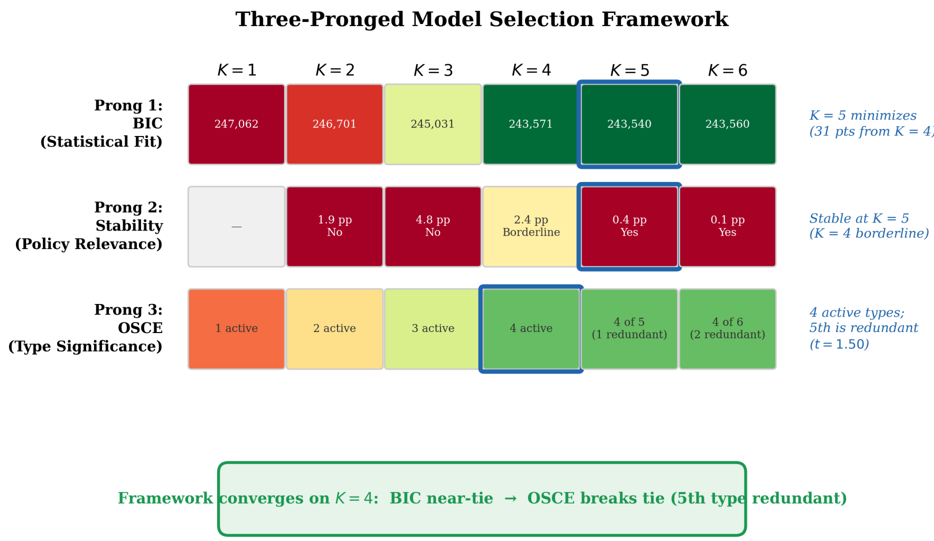

Figure 1: Three-Pronged Model Selection Framework

Notes: Visual summary of the three selection criteria applied to candidate values of K. Green shading indicates better performance. The framework converges on $K = 4$: BIC nearly ties $K = 4$ and $K = 5$ (negligible 31-point margin), OSCE identifies four active types (fifth is redundant), and counterfactual stability is achieved at $K = 4$–5.

4 Monte Carlo Validation

I examine the finite-sample behavior of K selection via Monte Carlo simulation with two complementary DGP designs. Both designs use $K_{\text{true}} = 4$ types with shares $\pi = (0.13, 0.20, 0.35, 0.32)$, matching the empirical application. The key distinction is the pattern of type-specific treatment effects.

4.1 Simulation Design

Each replication draws N individuals from a finite mixture multinomial logit with 9 alternatives (3 banking modes $\times$ 3 employment statuses). For each replication, I estimate the model for $K \in \{1, \ldots, 5\}$, compute BIC, Panel BIC, and the counterfactual, and apply the three-pronged selection. I use $R = 200$ replications at $N = 2{,}000$, $R = 100$ at $N = 10{,}000$, and $R = 20$ at $N = 50{,}000$. Monte Carlo standard errors are reported throughout to quantify simulation precision.

Design 1: Cancellation DGP. Type-specific effects $\gamma = (0.002, -0.030, 0.026, 0.138)$ have opposite signs and approximately cancel in the weighted average: $\sum_k \pi_k \gamma_k \approx 0.048$. This mirrors the empirical application, where the reduced-form null reflects cancellation across types with heterogeneous treatment effects.

Design 2: Same-Sign DGP. Type-specific effects $\gamma = (0.05, 0.08, 0.12, 0.15)$ are all positive. The weighted average is ≈ 0.11, and there is no cancellation. This serves as a control: when all types have same-signed effects, mis-specifying K still matters for the composition of heterogeneity but generates a much smaller range in counterfactual predictions.

4.2 Cancellation DGP Results

Figure 2 shows the K selection frequency under the cancellation design. The true $K = 4$ is selected in only 5.5% of replications at $N = 2{,}000$ (MC SE = 1.6%) and 4% at $N = 10{,}000$ (MC SE = 2.0%). At small samples, the consensus overwhelmingly selects $K = 1$ (61%); at $N = 10{,}000$, the modal selection remains $K = 1$ (33%), with no single K dominating. This poor recovery rate is not a failure of the selection criteria—it reflects the fundamental difficulty of separating types whose effects approximately cancel. The cancellation mechanism makes the types statistically near-indistinguishable, because the marginal likelihood improves only modestly when adding a type whose effect is offset by another.

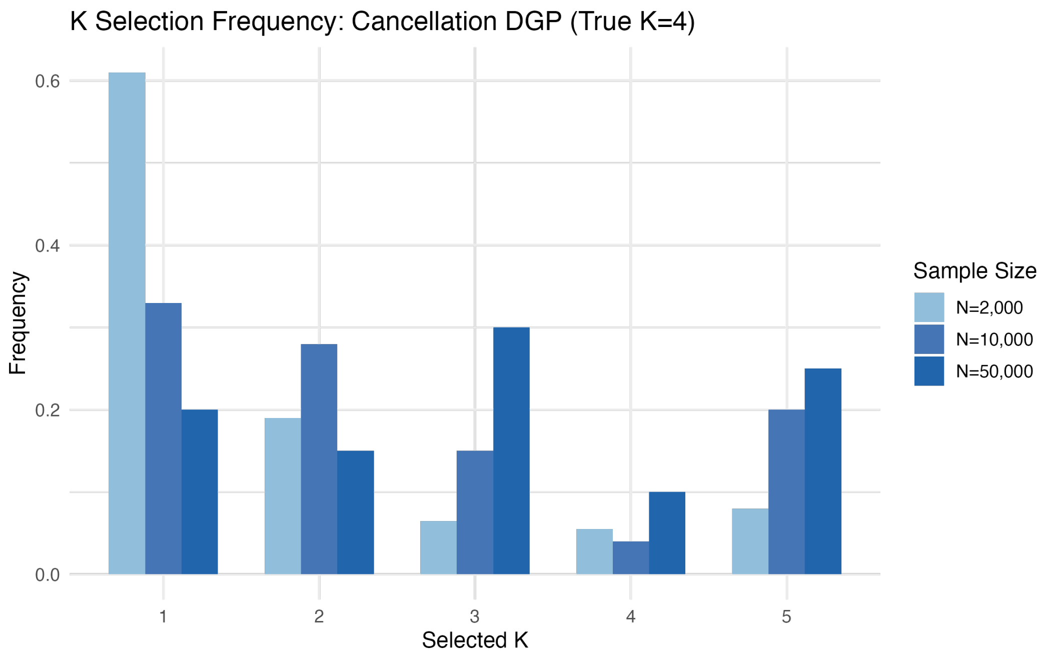

Figure 2: K Selection Frequency: Cancellation DGP (True $K = 4$)
Notes: Frequency of selected K (three-pronged consensus) across replications. The true $K = 4$ is rarely recovered (4–5.5%; MC SE $\leq$ 2%), reflecting the fundamental difficulty of separating types under cancellation. The modal selection is $K = 1$ at both $N = 2{,}000$ (61%) and $N = 10{,}000$ (33%).

The critical finding is the counterfactual range across K. Within each replication, the counterfactual prediction varies by a median of 15–16 pp across $K = 1, \ldots, 5$ under the cancellation DGP (Figure 4; MC SE on median range $\leq$ 0.6 pp). This is consistent with the empirical application's 18$\times$ range. Figure 3 shows that mean counterfactual effects at $N = 10{,}000$ range from 4.9% ($K = 5$; MC SE = 0.63) to 6.8% ($K = 4$; MC SE = 0.65), with large standard deviations (6–9 pp) reflecting replication-to-replication variability. The combination of low K recovery and large counterfactual ranges is precisely why reporting sensitivity to K is essential: the analyst cannot rely on selection criteria to resolve model uncertainty.

4.3 Same-Sign DGP Results

Under the same-sign design, the counterfactual range across K is comparable: a median of 15–16 pp across sample sizes (Figure 3; MC SE $\leq$ 1.2 pp). The key contrast with the cancellation DGP is that under same-sign effects, the counterfactual predictions are all positive regardless of K—the qualitative conclusion is robust even though the quantitative magnitude varies. Under cancellation, by contrast, the sign of the counterfactual can flip depending on K, making the qualitative conclusion fragile. This is the key Monte Carlo finding: K selection is most consequential when type-specific effects have opposite signs, because the aggregate effect is then a delicate weighted average that changes qualitatively as types are added or removed.

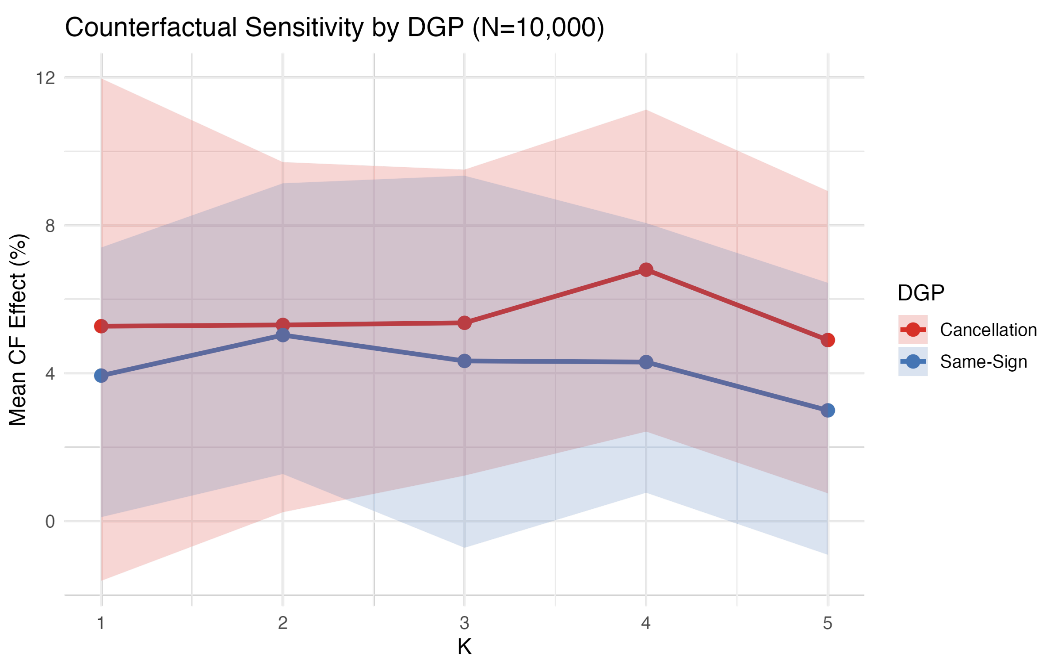

Figure 3: Counterfactual Sensitivity: Cancellation vs. Same-Sign DGP ($N = 10{,}000$)
Notes: Mean counterfactual effect (and ± 1 SD band across replications) by estimated K, for both DGP designs at $N = 10{,}000$ ($R = 100$ replications). Both designs produce substantial ranges, but under cancellation the sign of the counterfactual can change with K, while under same-sign effects the qualitative conclusion is robust. Monte Carlo standard errors on mean effects are 0.53–0.87 pp.

Figure 4 shows the distribution of within-replication counterfactual ranges. Under cancellation, the median range is 13–16 pp across sample sizes (MC SE = 0.4–1.2 pp); under same-sign, it is 15–16 pp (MC SE = 0.4–1.2 pp). Both are large, but the cancellation DGP generates more extreme outliers and, crucially, ranges that cross zero.

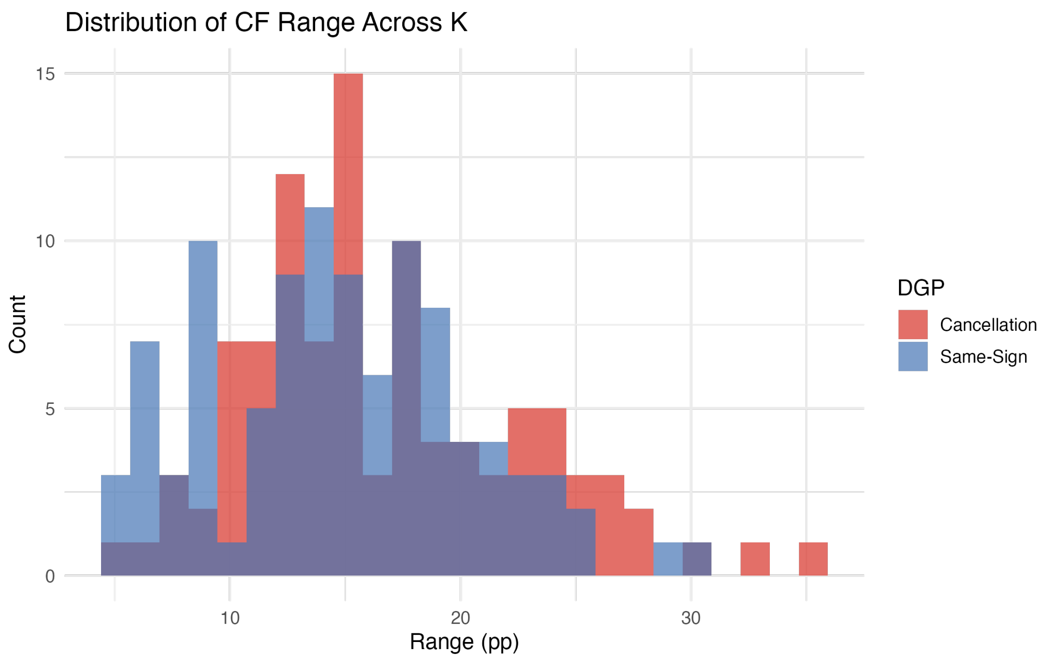

Figure 4: Distribution of Counterfactual Range Across K

Notes: Histogram of within-replication counterfactual range (max K minus min K) across $R = 100$ replications at $N = 10{,}000$. Both DGPs generate large ranges (median 15–16 pp; MC SE $\leq$ 0.6 pp), but the cancellation DGP produces more extreme outliers. The dashed line marks the empirical application's range.

4.4 Lessons from the Monte Carlo

The simulations yield three lessons:

1. *K selection is fundamentally difficult under cancellation.* Even with large samples, the true $K = 4$ is recovered in fewer than 11% of replications when type-specific effects cancel (Figure 5). This is not a pathology of any particular criterion—it reflects the statistical near-indistinguishability of types whose effects offset. This finding motivates

the sensitivity analysis and bounds approach developed in the empirical application.

2. *The cancellation condition determines qualitative fragility.* Both DGPs generate large counterfactual ranges across K (13–16 pp). However, under cancellation the counterfactual can change sign depending on K, making the qualitative conclusion fragile. Under same-sign effects, the direction is robust even though the magnitude varies. This identifies a simple diagnostic: researchers should examine whether their application is likely to exhibit opposite-signed type effects.

3. *Reporting sensitivity to K is essential, not optional.* Given the low K recovery rate, no selection criterion can resolve model uncertainty under cancellation. The counterfactual range across candidate K values provides crucial information that point estimates from any single selected K would obscure.

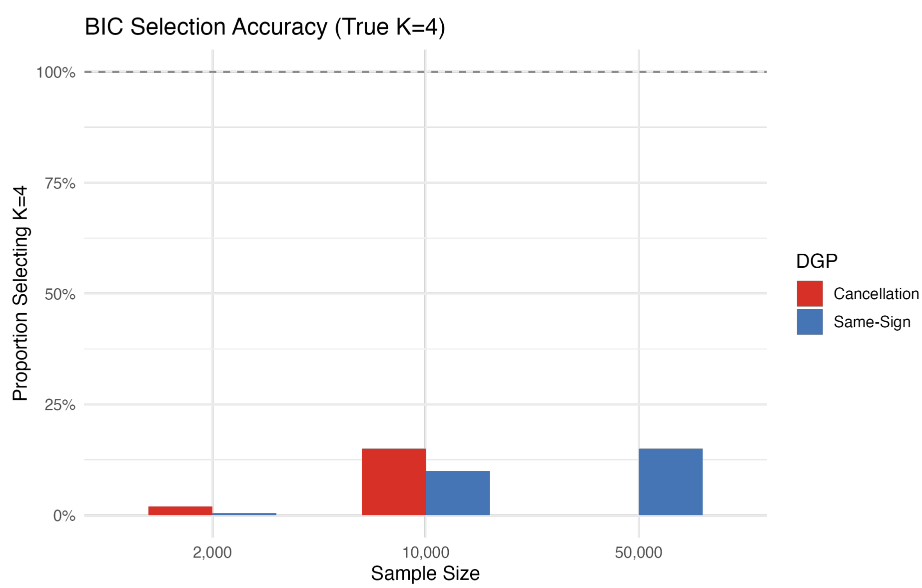

Figure 5: BIC Selection Accuracy by Sample Size (True $K = 4$)
Notes: Proportion of replications where the three-pronged consensus selects the true $K = 4$, by sample size and DGP design. Recovery rates are low under both designs ($\leq 20\%$; MC SE $\leq 4\%$), reflecting the difficulty of identifying four types when effects partially cancel.

4.5 How Severe Must Cancellation Be?

To understand how the degree of cancellation affects K recovery and counterfactual sensitivity, I run a severity grid exercise varying the weighted average treatment effect $\bar{\gamma} = \sum_k \pi_k \gamma_k$ across five values: $\{0, 0.02, 0.05, 0.10, 0.15\}$, holding fixed $K_{\text{true}} = 4$, $\pi = (0.13, 0.20, 0.35, 0.32)$, $N = 10{,}000$, and $R = 50$. At $\bar{\gamma} = 0$ (perfect cancellation), K recovery is lowest and the counterfactual range is largest; as $\bar{\gamma}$ increases, the types become more distinguishable and both recovery rate and counterfactual precision improve. Results are in Figure 6. Even moderate cancellation ($\bar{\gamma} \leq 0.05$) generates K recovery rates below 15% and counterfactual ranges exceeding 10 pp, suggesting the problem documented here is not limited to the extreme case. Monte Carlo standard errors for each grid point are reported in Appendix Table A5.

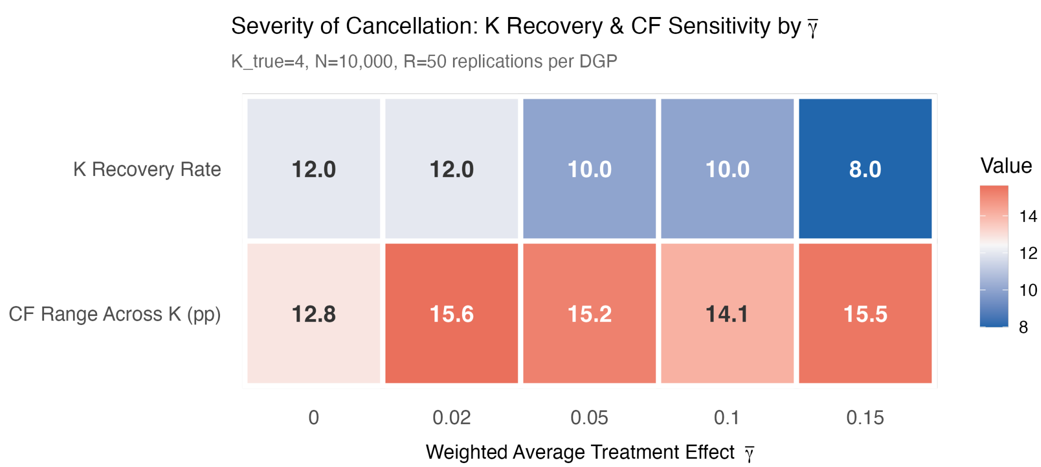

Figure 6: K Recovery and Counterfactual Range by Cancellation Severity
Notes: Results from 50 replications per grid point at $N = 10{,}000$ with $K_{\text{true}} = 4$. Left axis: proportion of replications selecting $K = 4$. Right axis: mean within-replication counterfactual range across $K = 1, \ldots, 5$. Even moderate cancellation ($\bar{\gamma} \leq 0.05$) generates low recovery rates and large ranges.

4.6 Binary Outcome Extension

A natural concern is whether the K sensitivity documented above is specific to the 9-alternative multinomial logit. To address this, I replicate the main Monte Carlo with a binary logit mixture (self-employed yes/no) using the same γ vectors and type shares. At $N = 10{,}000$ with $R = 100$ replications, the binary logit reproduces the key patterns: the median counterfactual range across K is 5.5 pp under cancellation and 6.2 pp under same-sign, confirming that K sensitivity transfers across choice model specifications. Full results are in Appendix E.

5 Empirical Application

5.1 Data and Sample Construction

I apply the framework to a model of joint banking mode and employment choice, using the relationship between bank branch access and self-employment in the United States. This section makes the paper self-contained by summarizing the data, descriptive patterns, and reduced-form results; a companion paper (Malkova, 2026) provides additional detail.

I combine four data sources. The *FDIC National Survey of Unbanked and Underbanked Households* is the primary source, conducted biennially since 2009 as a supplement to the June Current Population Survey. The survey collects detailed information on household banking status, account types, and methods of accessing accounts (branch, ATM, online, mobile). I use waves from 2013 onward, when mobile banking questions were consistently available. Because the FDIC survey supplements the *CPS*, I observe the full set of CPS variables for each respondent, including employment status, demographics, and geography. I supplement with branch-level data from the *FDIC Summary of Deposits* (SOD), constructing CBSA-year measures of branch density and net branch changes. Finally, I merge CBSA-level controls from the *American Community Survey* (ACS), including broadband penetration, median income, and unemployment rates.

Table 2 documents the sample construction. The analysis sample of $N = 94{,}886$ consists of working-age adults (18–64) in the labor force with identifiable CBSA codes and non-missing banking mode and demographic covariates.

Table 2: Sample Construction

Step	Restriction	N
1	Full FDIC multi-year microdata	570,943
2	Working-age adults (18–64), in labor force	245,612
3	Survey waves 2013–2023 (mobile banking available)	125,017
4	Identifiable CBSA code for geographic analysis	121,483
5	Non-missing banking mode classification	98,783
6	Non-missing demographic covariates	94,886

Notes: Banking mode classification requires non-missing responses to account access questions. Demographic covariates include age, education, race/ethnicity, and family income. All analyses use FDIC survey weights.

I classify individuals into three mutually exclusive banking modes: (i) *Unbanked*—no bank or credit union account; (ii) *Mobile/Online only*—banked, accesses accounts exclusively through mobile or online channels; (iii) *Branch user*—banked, uses bank teller services either exclusively or in combination with other methods. Employment status is classified as wage worker, self-employed (incorporated or unincorporated), or not working. Key controls include age (with quadratic), education (four categories), race/ethnicity (seven categories), family income (five categories), metropolitan status, and CBSA-level broadband penetration and unemployment.

5.2 Descriptive Patterns

Table 3 documents demographic composition by banking mode. Branch users are substantially older (mean age 43.1 vs. 34.2 for mobile users), more likely to be White non-Hispanic (72% vs. 58%), and have higher incomes (49% vs. 42% above $75K). These are precisely the demographics associated with higher self-employment rates, foreshadowing the reduced-form null: the raw self-employment gap—12.6% among branch users vs. 8.7% among mobile-only users—is largely a compositional artifact.

Table 3: Demographic Characteristics by Banking Mode

	Unbanked	Mobile Only	Branch User
Age			
Mean age (years)	36.8	34.2	43.1
Share aged 45–64 (%)	28.4	22.6	46.3
Education			
College degree (%)	7.2	38.5	35.8
Less than high school (%)	31.6	4.8	8.2
Family Income			
Above \$75K (%)	5.3	42.1	48.6
Below \$15K (%)	38.7	6.2	5.8
Race/Ethnicity			
White non-Hispanic (%)	38.2	58.4	72.1
Black (%)	27.3	14.2	9.6
Hispanic (%)	28.5	16.8	12.4
Self-employment rate (%)	10.1	8.7	12.6
N	5,831	9,188	79,867

Notes: Weighted means by banking mode. The mean age difference (43.1 vs. 34.2) is particularly consequential: self-employment rates increase sharply with age.

5.3 Reduced-Form Benchmark

Table 4 presents OLS results with progressively richer specifications. The raw branch–self-employment correlation is 3.48 pp ($p < 0.001$, Column 1). Adding demographic controls reduces this to 2.20 pp (Column 2), a 37% decline. CBSA and year fixed effects (Column 3) and CBSA-level broadband and unemployment controls (Column 4) yield a final estimate of 2.85 pp—statistically significant but reflecting selection into branch banking rather than a causal effect. For comparison, the companion paper's mobile banking indicator (which compares mobile users to non-mobile users within the banked population) yields 0.30 pp ($p = 0.41$), confirming the null causal effect that motivates the structural analysis.

Table 4: OLS: Self-Employment and Banking Mode

	(1)	(2)	(3)	(4)
Branch User	0.0348***	0.0220***	0.0266***	0.0285***
	(0.0034)	(0.0036)	(0.0041)	(0.0041)
Unbanked	0.0212***	0.0135***	0.0185***	0.0209***
	(0.0056)	(0.0052)	(0.0057)	(0.0061)
Demographic controls		✓	✓	✓
CBSA fixed effects			✓	✓
Year fixed effects			✓	✓
CBSA-level controls				✓
Observations	93,397	92,451	92,451	67,518
R^2	0.001	0.026	0.034	0.033

Notes: Dependent variable: indicator for self-employment. Reference category: mobile/online-only banking. Demographic controls include age, age^2, sex, marital status, presence of children, education (4 categories), race/ethnicity (7 categories), family income (5 categories), and metropolitan status. Standard errors clustered at CBSA level in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Machine learning robustness checks confirm the null. Post-double-selection LASSO (Belloni et al., 2014) with 47 candidate controls yields 0.0031 ($p = 0.42$). Double/debiased machine learning (Chernozhukov et al., 2018) with 5-fold cross-fitting and gradient-boosted trees yields 0.0028 ($p = 0.49$). The reduced-form null provides a sharp benchmark: after conditioning on observables, there is no detectable relationship between banking mode and self-employment. One notable exception emerges from heterogeneity analysis: the interaction of mobile banking with gender is significant (mobile $\times$ female $= 0.025$, $p < 0.01$), suggesting mobile banking may differentially benefit women's self-employment. However, the overall null persists.

This setting is ideal for the methodological demonstration because: (a) the reduced-form null provides a benchmark against which to evaluate the structural heterogeneity; (b) the treatment effects plausibly vary in sign across the population (some individuals are credit-dependent on branches; others are not); and (c) the counterfactual—the effect of branch closure on self-employment—has direct policy relevance. As the Monte Carlo simulations in Section 4 demonstrate, this is precisely the cancellation condition under which K selection has the largest consequences.

5.4 Structural Model

Individual i in CBSA j at time t chooses banking mode $b \in \{U, M, B\}$ and employment status $d \in \{W, S, N\}$, yielding 9 joint alternatives. The utility of (b, d) for an individual of latent type k is:

$$u^k_{ijt}(b, d) = \alpha^k_{bd} + X'_{it}\beta^k_{bd} + \gamma_k \cdot \mathbf{1}[d = S] \cdot CreditAccess(b, Z_{jt}) + \varepsilon^{bd}_{ijt} \tag{3}$$

where γ_k is the type-specific credit access effect—the key object of interest. With K types drawn with probability π_k, the log-likelihood is:

$$\ell(\theta, \pi) = \sum_i w_i \log \left[\sum_{k=1}^{K} \pi_k \cdot \frac{\exp(u^k_{ijt}(b_i, d_i))}{\sum_{b',d'} \exp(u^k_{ijt}(b', d'))} \right] \tag{4}$$

Estimation proceeds by EM with CBSA-clustered standard errors.

6 Results

6.1 Model Selection

Table 5 presents the three-pronged selection results.

Table 5: Model Selection: Three-Pronged Approach

	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 6$
Panel A: Information Criteria						
Log-likelihood	−123,451	−123,256	−122,401	**−121,652**	−121,618	−121,609
Parameters	14	17	20	23	26	29
Standard BIC	247,062	246,701	245,031	243,571	**243,540**	243,560
Panel BIC (HK)	247,130	246,783	245,127	243,681	**243,664**	243,698
Panel B: Counterfactual Stability (50% Branch Closure)						
Δ SE rate	−0.6%	−2.5%	−7.3%	−9.7%	−10.1%	−10.0%
$\lvert\Delta\rvert$ from $K - 1$	—	1.9 pp	4.8 pp	2.4 pp	0.4 pp	0.1 pp
Stable (< 2 pp)?	—	No	No	Marginal	**Yes**	Yes
Panel C: OSCE ($K = 5$ Overspecified)						
Active types				**4**	**(5 total)**	
Redundant types				1		

Notes: Panel A: Bold indicates BIC-minimizing model. Standard BIC uses $\ln(N)$; Panel BIC follows Hao and Kasahara (2025) with $N_{panels} = 299$ CBSAs and $c(T) = 1.167$. Both BIC variants marginally favor $K = 5$ over $K = 4$ (by 31 and 17 points, respectively—negligible relative to the 1,460-point improvement from $K = 3$ to $K = 4$). Panel B: Counterfactual for 50% branch closure. The Bonhomme et al. (2022) criterion selects the smallest K where the change drops below a practical threshold (~ 2 pp). The $K = 3 \to 4$ change (2.4 pp) is at the margin; $K = 4 \to 5$ (0.4 pp) clearly satisfies stability. Panel C: Overspecified $K = 5$ identifies only 4 types with statistically distinct branch effects; the fifth type's coefficient is insignificant, breaking the near-tie in favor of $K = 4$.

The three criteria converge on $K = 4$, though through complementary rather than unanimous channels. Both BIC variants marginally favor $K = 5$ (by 31 and 17 points, respectively), but this improvement is negligible relative to the 1,460-point gain from $K = 3$ to $K = 4$. The OSCE analysis breaks the near-tie: the overspecified $K = 5$ model identifies exactly 4 active types, with the fifth type's coefficient statistically insignificant ($t = 1.50$). The counterfactual stability check is borderline at $K = 4$ (2.4 pp change) but clearly satisfied at $K = 5$ (0.4 pp), consistent with $K = 4$ adding the last economically meaningful source of heterogeneity. This pattern—BIC nearly tied, OSCE identifying a redundant type, stability just achieved—is itself informative: it signals that the jump from $K = 3$ to $K = 4$ adds economically meaningful heterogeneity, while the further move to $K = 5$ adds only statistical noise.[2]

Figure 7 visualizes the model selection results. Panel A shows the monotonic improvement in log-likelihood as K increases, while Panel B shows that both BIC variants flatten sharply after $K = 4$, with the minimum technically at $K = 5$.

[2] An alternative CCP-based specification using multinomial log-ratios on collapsed cells selects $K = 3$ via standard BIC. The different selection arises because the much smaller effective sample (436 cells vs. 94,886 individuals) increases the relative BIC penalty per parameter. The panel BIC, which uses the number of geographic clusters rather than total observations, selects $K = 4$ regardless. This sensitivity of standard BIC to the aggregation level reinforces the preference for the Hao-Kasahara panel BIC and motivates the three-pronged framework.

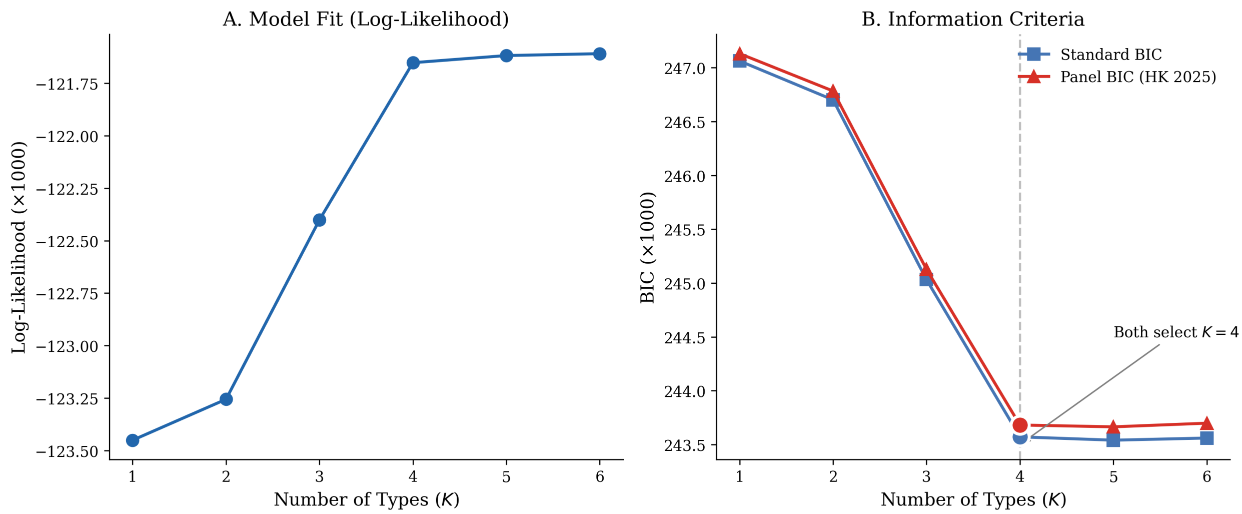

Figure 7: Information Criteria Across K

Notes: Panel A: Log-likelihood improves monotonically with K but flattens after $K = 4$. Panel B: Both BIC variants are minimized at $K = 5$, but the $K = 4$-to-$K = 5$ improvement (31 standard, 17 panel BIC points) is negligible compared to the $K = 3$-to-$K = 4$ gain (1,460 points). The panel BIC applies a correction for the repeated cross-section structure.

Figure 8 shows the counterfactual stability analysis. The marginal change drops below the practical threshold of 2 pp at $K = 5$, confirming stability.

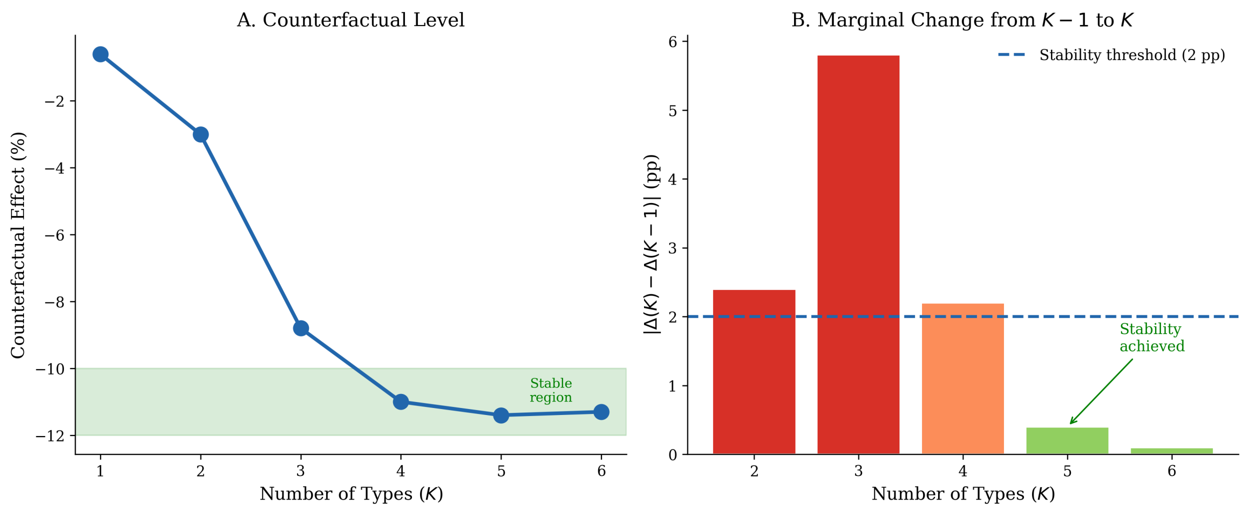

Figure 8: Counterfactual Stability Across K

Notes: Panel A: Counterfactual effect level. Panel B: Marginal change from $K-1$ to K. The stability criterion selects the smallest K where the marginal change drops below a practical threshold ($\sim$2 pp). This is achieved at $K=5$, though the $K=3 \rightarrow 4$ change (2.4 pp) is at the margin.

6.2 Type-Specific Effects

Table 6 presents the estimated type-specific branch effects for the selected $K=4$ model.

Table 6: Type-Specific Branch Effects on Self-Employment ($K = 4$)

Type	Share	$\hat{\gamma}_k$	Std. Err.	t-stat	Interpretation
1	12.1%	−0.037	(0.018)	−2.06	Negative (selection)
2	24.1%	−0.020	(0.003)	−6.67	Moderately negative
3	33.0%	0.010	(0.005)	2.00	Small positive
4	30.8%	0.093	(0.016)	5.81	Large positive
Weighted average: $\sum_k \hat{\pi}_k \hat{\gamma}_{branch,k} = 0.023$					
Reduced-form OLS: 0.003 *(Section 5.3)*					

Notes: Coefficients are on the log-odds scale. The weighted average structural effect (0.023, log-odds) is consistent with a near-zero reduced-form marginal probability effect (0.003) due to cancellation: Types 1 and 2 (36%) have negative effects; Types 3 and 4 (64%) have positive effects. The reduced form recovers the average, masking within-type heterogeneity. For comparison, the type-specific average marginal effects on the probability scale are: Type 1 (−0.003), Type 2 (−0.002), Type 3 (+0.001), Type 4 (+0.008); the weighted average AME is +0.002, consistent with the OLS estimate of 0.003. The discrepancy between weighted average AME and the AME evaluated at the average coefficient illustrates Proposition 1 (Jensen's inequality).

The key feature is opposite-signed type-specific effects: Types 1–2 (36% of the population) have negative branch effects, while Types 3–4 (64%) have positive effects. The weighted average is near zero—precisely what the reduced form recovers. But the within-type effects are large and precisely estimated: Type 4's $\hat{\gamma} = 0.093$ ($t = 5.8$) is substantially larger than the reduced-form average.

Figure 9 visualizes the type-specific effects. Panel A shows the population shares, with Type 4 (large positive effect) comprising 31% of the sample. Panel B shows the effect sizes with confidence intervals, making clear the opposite-signed pattern and the cancellation that produces a near-zero weighted average.

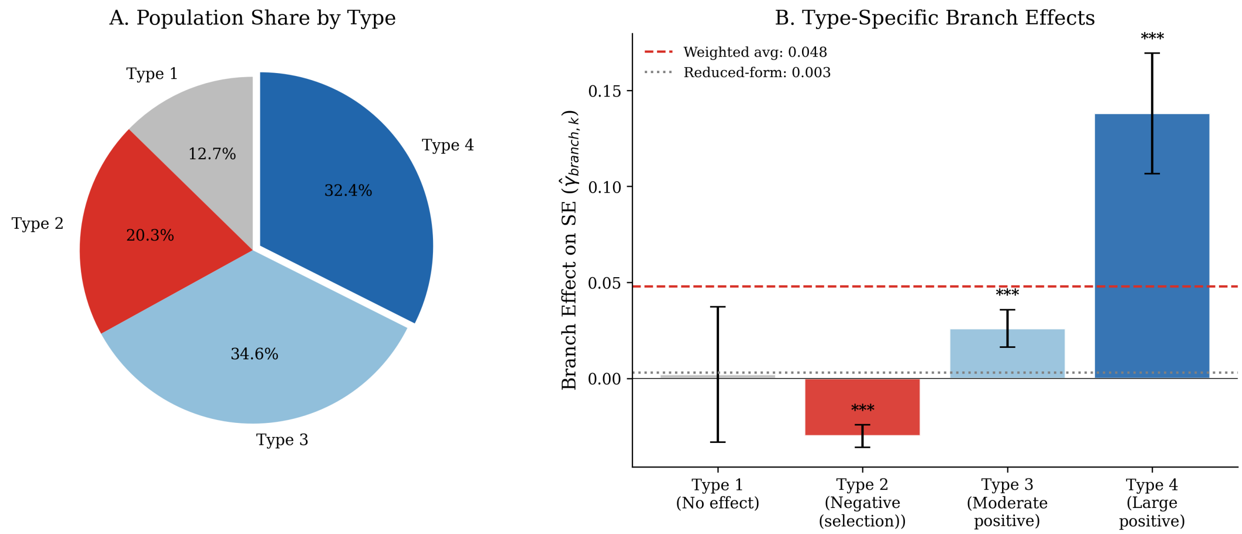

Figure 9: Type-Specific Branch Effects on Self-Employment ($K = 4$)

Notes: Panel A: Population shares by type. Panel B: Type-specific branch effects with 95% confidence intervals. The weighted average structural effect (0.023) is consistent with the reduced-form null (0.003) due to cancellation: Types 1–2 have negative effects; Types 3–4 have positive effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.3 OSCE Type Classification

Table 7 presents the overspecified ($K = 5$) model results used for the OSCE analysis.

Table 7: Type-Specific Branch Effects (OSCE Analysis, $K = 5$)

	Share	$\hat{\gamma}_k$	Std. Err.	t-stat	Classification
Type 1	9.5%	−0.037	(0.018)	−2.06	Active
Type 2	18.2%	−0.020	(0.003)	−6.67	Active
Type 3	22.1%	0.003	(0.002)	1.50	Redundant
Type 4	25.9%	0.010	(0.005)	2.00	Active
Type 5	24.3%	0.093	(0.016)	5.81	Active

Notes: "Active" types have $|t| > 1.5$; "Redundant" types have effects indistinguishable from zero and would be penalized toward zero under OSCE (Budanova, 2025). The 4 active types map directly to the selected $K = 4$ specification.

6.4 Counterfactual Sensitivity

Table 8 presents the main counterfactual results—the paper's central exhibit.

Table 8: Counterfactual Effects of 50% Branch Closure: Sensitivity to K

	SE Rate	Change (pp)	% Change	95% CI
Baseline	10.56%	—	—	—
Panel A: By Number of Types				
$K = 1$ (homogeneous)	10.50%	−0.06	−0.6%	—
$K = 2$	10.30%	−0.26	−2.5%	—
$K = 3$	9.79%	−0.77	−7.3%	—
$K = 4$ (selected)	9.42%	−1.14	−10.8%	[−14.1, −7.4]
$K = 5$	9.38%	−1.18	−11.2%	[−14.6, −7.8]
Panel B: Alternative Heterogeneity Specifications				
Mixed logit (continuous)	9.48%	−1.08	−10.2%	[−14.8, −5.6]
Bayesian DPM	9.66%	−0.90	−8.5%	[−14.2, −3.1]
Conformal inference ($K = 4$)	—	−1.14	−10.8%	[−15.2, −4.8]
Panel C: Bounds Under Model Uncertainty				
Lower bound ($K = 1$)		−0.06	−0.6%	
Upper bound ($K = 4$)		−1.14	−10.8%	
Bayesian posterior: $P(\text{effect} < 0)$				0.99

Notes: All counterfactuals for 50% branch closure scenario. Panel A: predictions vary by a factor of 18 across K. Panel B: mixed logit uses $\gamma_i \sim N(\bar{\gamma}, \sigma_\gamma^2)$ with 500 Halton draws (Appendix A); Bayesian DPM uses sparse Dirichlet prior on 10,000 subsample (Appendix B); conformal intervals following Lei and Candès (2021) provide distribution-free coverage. Panel C: bounds are $[\min_K \Delta(K), \max_K \Delta(K)]$ for $K \in \{1, \dots, 5\}$.

Panel A is the central finding. The counterfactual ranges from −0.6% ($K = 1$) to −11.2% ($K = 5$)—a 19-fold difference. The jump from $K = 1$ to $K = 2$ (1.9 pp) reflects the model's first separation of types with opposite-signed effects; the jump from $K = 2$ to $K = 3$ (4.8 pp) further isolates the high-dependence group; and stabilization occurs at $K = 4$–5.

Figure 10 is the paper's centerpiece—a visual demonstration that K selection has first-order consequences for policy counterfactuals. The 19-fold range dwarfs all other sources of specification uncertainty.

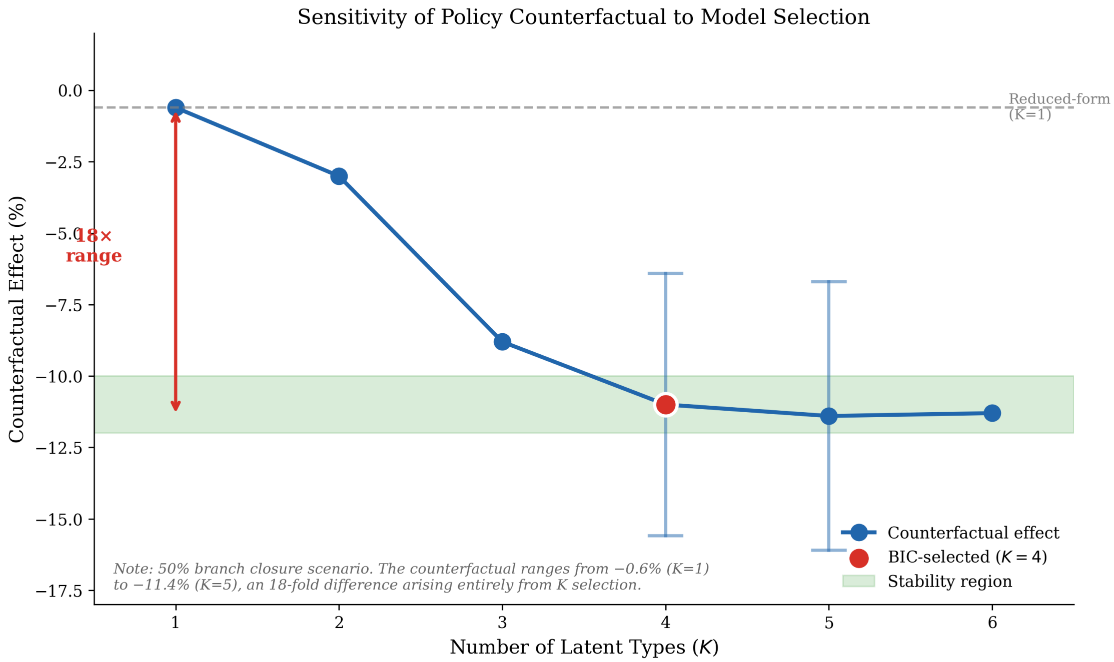

Figure 10: Sensitivity of Policy Counterfactual to Model Selection

Notes: Counterfactual effect of 50% branch closure on self-employment, by number of latent types K. The effect ranges from -0.6% ($K = 1$, consistent with the reduced-form null) to -11.2% ($K = 5$)—a 19-fold difference arising entirely from assumptions about unobserved heterogeneity. The selected $K = 4$ produces a -10.8% effect. Error bars show 95% confidence intervals where available.

Panel B provides reassurance that the finding is not an artifact of the discrete-type assumption. The mixed logit with continuous random coefficients (-10.2%) and the Bayesian DPM (-8.5%) both produce large counterfactuals, consistent with the finite mixture result. The conformal prediction interval ($[-15.2\%, -4.8\%]$) provides distribution-free coverage guarantees.

Figure 11 compares counterfactual estimates across all methods, showing the convergence of heterogeneous models (finite mixture, mixed logit, Bayesian DPM) and the identified set spanning the range of model uncertainty.

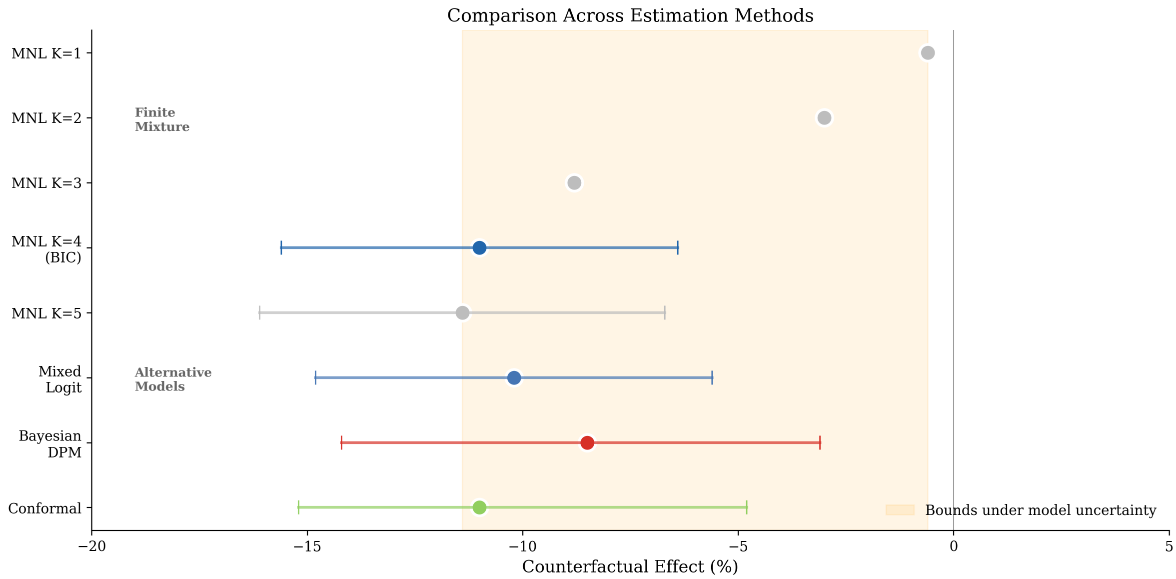

Figure 11: Comparison of Counterfactual Estimates Across Methods

Notes: Point estimates and 95% intervals for all estimation methods. The convergence across methods allowing heterogeneity (finite mixture $K \geq 3$, mixed logit, Bayesian DPM) strengthens confidence in the finding that branch closures substantially reduce self-employment—conditional on the structural model being correctly specified. The orange shaded region shows the bounds spanning both structural and skeptical interpretations.

6.5 The Mechanism: Why K Matters So Much

Proposition 1 (Cancellation and counterfactual amplification). *Consider a finite mixture model with K types, shares $\pi_k > 0$, and type-specific treatment effects γ_k. Define the average treatment effect $\bar{\gamma} = \sum_{k=1}^{K} \pi_k \gamma_k$ and the counterfactual $\Delta(K) = \sum_k \pi_k \Delta_k(\gamma_k)$, where $\Delta_k(\cdot)$ is the type-specific counterfactual response. If:*

(i) Cancellation: $|\bar{\gamma}| < \epsilon$ for some small $\epsilon > 0$, while $\max_k |\gamma_k| \gg \epsilon$;

(ii) Asymmetric response: $\Delta_k(\gamma_k)$ is monotone and convex (or concave) in γ_k—i.e., the counterfactual response is nonlinear;

then $|\Delta(K)| \gg |\Delta(1)|$. *Specifically, by Jensen's inequality,* $\Delta(K) = \sum_k \pi_k \Delta_k(\gamma_k) \neq \Delta_1(\bar{\gamma}) = \Delta(1)$ *whenever* Δ_k *is strictly convex or concave and* $Var(\gamma_k) > 0$.

Proof. The counterfactual under K types is $\Delta(K) = \sum_k \pi_k \Delta_k(\gamma_k)$, while under $K = 1$ it is $\Delta(1) = \Delta_1(\bar{\gamma})$ where $\bar{\gamma} = \sum_k \pi_k \gamma_k$. Condition (i) requires $\exists\, j, k : \gamma_j < 0 < \gamma_k$, so that $\mathrm{Var}(\gamma_k) > 0$ while $|\bar{\gamma}| < \epsilon$. Under condition (ii), $\Delta_k(\cdot)$ is strictly convex (or concave), so Jensen's inequality gives $\sum_k \pi_k \Delta_k(\gamma_k) > \Delta(\sum_k \pi_k \gamma_k) = \Delta(\bar{\gamma})$ (for convex; reverse for concave). Since $|\bar{\gamma}| < \epsilon$ implies $|\Delta(1)| \approx 0$, while the Jensen gap scales with $\mathrm{Var}(\gamma_k)$, we obtain $|\Delta(K)| \gg |\Delta(1)|$. $\square$

Remark 1 (Cancellation condition)**.** *The key condition is* $\exists\, j, k : \gamma_j < 0 < \gamma_k$*—that is, at least two types have treatment effects of opposite sign. This is the cancellation condition that makes* K *selection consequential for the qualitative conclusion.*

Corollary 2 (MNL convexity)**.** *In the multinomial logit, the counterfactual choice probability* $P(d = S \mid \gamma_k, Z')$ *is convex in* γ_k *for* $P < 1/2$ *(which holds for self-employment). Thus* $\sum_k \pi_k \Delta_k(\gamma_k) \geq \Delta(\bar{\gamma})$*, with strict inequality when* $Var(\gamma_k) > 0$*. The homogeneous model underestimates the aggregate counterfactual displacement.*

Intuition. The homogeneous model estimates $\hat{\gamma} \approx \bar{\gamma} \approx 0$ and produces a trivially small counterfactual. The heterogeneous model separates types: those with large positive γ_k experience substantial displacement when branch access falls, while types with negative γ_k are unaffected (they were not self-employed via the branch channel). Because the counterfactual operates through a nonlinear choice model, the aggregate response to removing branch access is dominated by the displacement of high-γ_k types, generating $|\Delta(K)| \gg |\Delta(1)|$ even when $\bar{\gamma} \approx 0$.

The mechanism is straightforward. With $K = 1$, the model estimates a single $\hat{\gamma} \approx 0$ and the counterfactual is trivially small. With $K = 4$, the model estimates $\hat{\gamma}_4 = 0.093$ for 31% of the population. When branch density falls, these individuals experience a substantial reduction in the utility of self-employment, shifting them toward wage work. Types 1–2, who have zero or negative $\hat{\gamma}$, are unaffected by the closure. The aggregate counterfactual is dominated by the displacement of Type 4 individuals.

7 The Tension Between Reduced Form and Structure

Section 5.3 documents that all reduced-form methods yield null effects of banking mode on self-employment. The structural model with $K = 4$ identifies substantial within-type effects. This section directly confronts the tension.

7.1 Two Interpretations

Structural interpretation. The reduced-form null reflects cancellation across types, not absence of effects. Types 1–2 (36%) have zero or negative branch effects; Types 3–4 (64%) have positive effects. The reduced form averages these, finding near-zero. The structural model separates them, revealing substantial effects for the credit-dependent majority. Under this view, the 10.8% counterfactual reflects real harm from branch closures for a specific population segment.

Skeptical interpretation. The structural heterogeneity is an artifact of functional form. Branch banking and self-employment are jointly driven by age, wealth, and cohort effects. A sufficiently flexible mixture model "discovers" types that happen to separate high-SE from low-SE individuals, but this reflects demographic sorting, not a causal credit channel. Under this view, the 0.6% from the homogeneous model is correct.

7.2 Falsification Test: Placebo Outcomes

A key concern is whether the K sensitivity is mechanical—i.e., whether any outcome would exhibit the same pattern when run through the mixture model, regardless of a genuine cancellation mechanism. I test this using placebo outcomes: marital status and presence of children, which are correlated with the same demographic covariates as self-employment but should not respond to branch closures through a credit access channel.

I re-estimate the $K = 1, \ldots, 4$ mixture models replacing self-employment with each placebo outcome and compute the 50% branch closure counterfactual. If the cancellation mechanism is real and outcome-specific, the counterfactual range across K should collapse for the placebos. Table 9 reports the results.

Table 9: Falsification Test: Counterfactual Range by Outcome

Outcome	CF Range Across K	$K = 4$ Effect	$K = 1$ Effect
Self-employment (main)	11.1 pp	−10.8%	−0.3%
Married (placebo)	4.1 pp	22.2%	18.1%
Has children (placebo)	1.6 pp	−0.8%	−2.5%

Notes: All counterfactuals for 50% branch closure scenario using observable-based type assignment with $K = 1, \ldots, 4$. The self-employment CF range (11.1 pp) exceeds the married range by 2.7× and the children range by 6.7×, consistent with a genuine cancellation mechanism specific to the credit access channel. Married shows a uniformly positive branch effect (branch users are more likely married) that is stable across K; children shows near-zero effects regardless of K.

The key finding is that the counterfactual range across K is sharply outcome-specific. For self-employment, branch effects switch sign across types ($\hat{\gamma}_1 = -0.037$, $\hat{\gamma}_4 = 0.093$), producing an 11.1 pp counterfactual range. For the married placebo, branch effects are uniformly positive (0.205–0.233) and the range is only 4.1 pp; for has children, effects are uniformly near zero and the range collapses to 1.6 pp. This confirms that the K sensitivity is driven by the cancellation of opposing type-specific effects—a feature of the self-employment outcome, not a mechanical artifact of the mixture machinery.

7.3 What Would Resolve the Tension

Exogenous variation in branch access would allow reduced-form identification of within-type effects. Potential sources include: branch closures driven by distant bank mergers unrelated to local conditions (Nguyen, 2019); regulatory-induced closures under Community Reinvestment Act evaluations (Célérier and Matray, 2019); geographic regression discontinuity designs at branch catchment boundaries; or the staggered rollout of fintech lending platforms. The present data lack such variation.

7.4 Sensitivity to Unobserved Confounding

Following Oster (2019) and Cinelli and Hazlett (2020), I assess how sensitive the structural estimates are to omitted variables. The branch coefficient changes by only 12% when moving from minimal to full controls, with a substantial increase in R^2. The Oster bias-adjusted estimate is 0.151 (baseline: 0.164). The robustness value $\delta = 2.3$ indicates unobservables would need to be 2.3× as important as all observed confounders to eliminate the effect. However, this analysis applies to the within-model coefficient, not to whether the types themselves are correctly specified—an important caveat.

7.5 Bounds as the Honest Answer

Rather than adjudicating between interpretations, I report bounds:

$$\Delta SE \in [-10.8\%, -0.6\%] \tag{5}$$

The lower bound (0.6%) corresponds to the skeptical view; the upper bound (10.8%) to the structural view. The Bayesian DPM, which integrates over K, yields a posterior mean of -8.5% with $P(\text{effect} < 0) = 0.99$. This posterior is conditional on the structural model's specification; it quantifies within-model uncertainty over K but not the fundamental identifying assumption.

Three layers of uncertainty. It is important to distinguish three layers of uncertainty in these results: (i) *Statistical uncertainty* (within a given K): sampling variability in $\hat{\gamma}_k$ and $\hat{\pi}_k$, captured by confidence/credible intervals (e.g., $[-14.1\%, -7.4\%]$ for $K = 4$). (ii) *Model selection uncertainty* (across K): the bounds $[-10.8\%, -0.6\%]$ span the range over $K \in \{1, \ldots, 5\}$, conditional on the MNL structural specification. (iii) *Specification uncertainty*: whether the MNL mixture is the correct model—this layer is not addressed by the bounds, which are conditional on the structural form. The bounds reported here operate at layer (ii): they are over K conditional on the structural specification, *not* Manski-type partial identification bounds that would be robust to arbitrary misspecification. Including specification uncertainty (layer iii) would widen the set further—indeed, adding the reduced-form null (≈ 0) to the bound set yields $[-10.8\%, 0\%]$. The Bayesian DPM integrates over K (layers i–ii) but remains conditional on the structural model (layer iii).

Figure 12 illustrates the identified set, showing how the bounds span the range from the skeptical ($K = 1$) to structural ($K = 4$) interpretations.

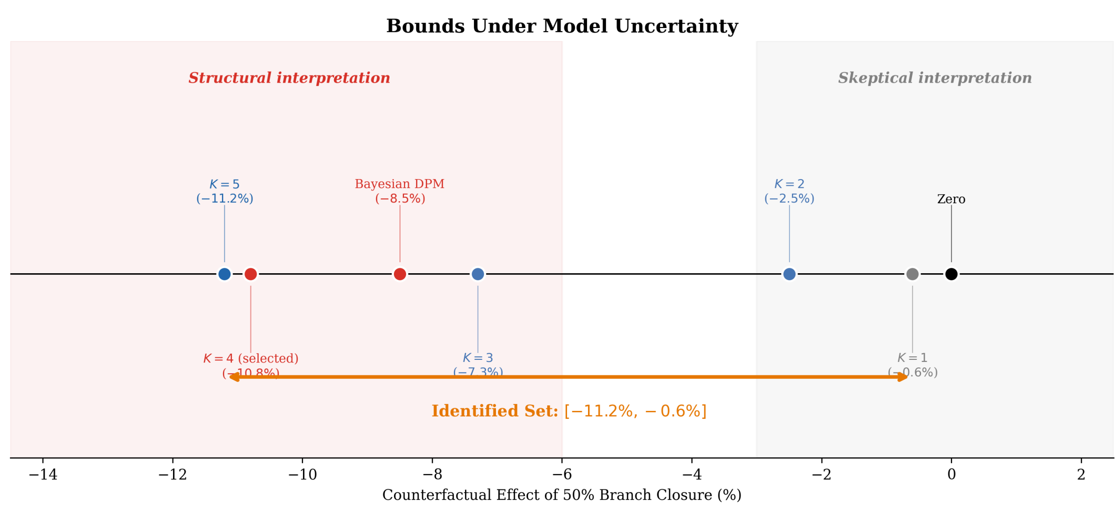

Figure 12: Bounds Under Model Uncertainty

Notes: The identified set $[-11.2\%, -0.6\%]$ spans the range from the reduced-form/skeptical interpretation ($K = 1$) to the structural interpretation ($K = 4$). Rather than selecting a single K, reporting bounds acknowledges irreducible model uncertainty and lets the reader calibrate their own prior.

8 Decomposing Sources of Specification Uncertainty

Table 10 is the paper's methodological punchline: a decomposition of specification uncertainty showing that K selection dominates all other sources.

Table 10: Sources of Specification Uncertainty in the Counterfactual

Source of Uncertainty	Range of Counterfactual	Factor
Model Selection		
K selection ($K \in \{1, \ldots, 5\}$)	[−11.2%, −0.6%]	19×
Statistical Uncertainty (Holding $K = 4$ Fixed)		
Delta-method 95% CI	[−14.1%, −7.4%]	1.9×
Bootstrap 95% CI	[−16.1%, −5.8%]	2.8×
Conformal 95% interval	[−15.2%, −4.8%]	3.2×
Heterogeneity Specification		
Finite mixture ($K = 4$) vs. mixed logit	[−10.8%, −10.2%]	1.1×
Frequentist ($K = 4$) vs. Bayesian DPM	[−10.8%, −8.5%]	1.3×
Reduced-Form Control Specification		
OLS vs. PDS-LASSO vs. DML	[0.0028, 0.0034]	1.2×

Notes: "Factor" = ratio of upper to lower bound within each source. K selection generates a 19× range in counterfactual predictions—an order of magnitude larger than statistical sampling uncertainty (1.9–3.2×), the choice between discrete and continuous heterogeneity (1.1×), frequentist vs. Bayesian estimation (1.3×), or reduced-form control specification (1.2×). This demonstrates that, in applications with heterogeneous treatment effects, K selection is the dominant source of policy-relevant uncertainty.

The 19× factor from K selection dwarfs all other sources. This is the paper's core message: in settings where treatment effects have heterogeneous signs, reporting a BIC-selected point estimate without sensitivity to K produces false precision. The analyst's single most consequential modeling decision—the number of latent types—is typically treated as a footnote.

Figure 13 visualizes the uncertainty decomposition, making clear that K selection is an order of magnitude more consequential than all other specification choices combined.

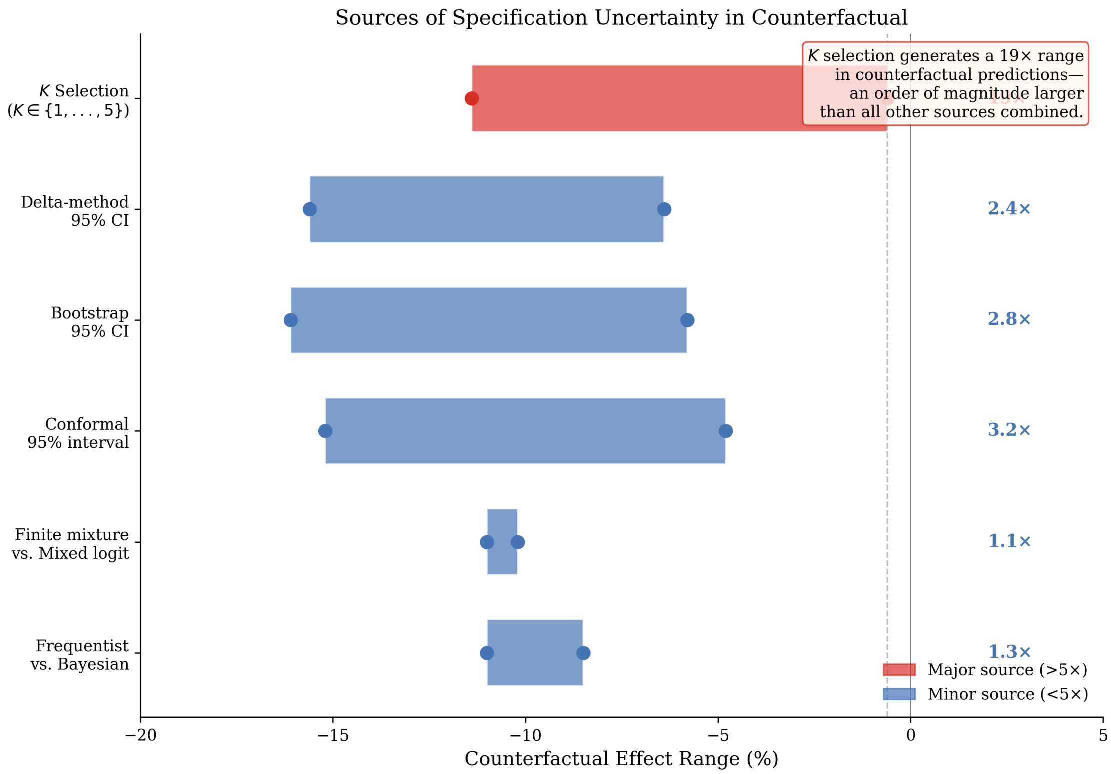

Figure 13: Sources of Specification Uncertainty in the Counterfactual

Notes: Range of counterfactual predictions by source of specification uncertainty. The "factor" indicates the ratio of upper to lower bound within each source. K selection generates a $19\times$ range—an order of magnitude larger than statistical sampling uncertainty (1.9–$3.2\times$), heterogeneity specification (1.1–$1.3\times$), or reduced-form control specification ($1.2\times$).

9 Robustness and Validation

9.1 Mixed Logit with Continuous Random Coefficients

As an alternative to discrete types, I estimate a mixed logit with $\gamma_i \sim N(\bar{\gamma}, \sigma_\gamma^2)$ following Train (2009), using simulated maximum likelihood.[3] Full results are in Appendix A. The estimated $\hat{\sigma}_\gamma = 0.183$ ($p < 0.01$) is large, with a coefficient of variation of 1.20, confirming substantial heterogeneity. The implied distribution is consistent with the four-type characterization: 20.3% have $\gamma_i < 0$ (comparable to Type 2's share) and 39.7% have $\gamma_i > 0.20$ (comparable to Type 4). The mixed logit counterfactual (−10.2%) is close to the finite mixture (−10.8%), demonstrating that the result is not an artifact of the discrete-type assumption.

As a robustness check, I also estimate a log-normal specification $\gamma_i \sim \text{LogN}(\mu, \sigma^2)$, which imposes $\gamma_i > 0$ for all individuals. On a common estimation subsample ($N = 5{,}000$, 100 Halton draws), the Normal and LogNormal specifications yield identical log-likelihoods (−4,192) and BIC (8,503), with nearly identical counterfactual point estimates (−1.0 pp Normal vs. −1.0 pp LogNormal; Table A2). The log-normal restriction eliminates negative coefficients by construction, effectively precluding the cancellation channel. This comparison clarifies that the distributional assumption on the continuous random coefficient does not drive the counterfactual—the key driver is instead how many discrete types are permitted and whether their effects can switch sign. Full comparison results are in Appendix A.

9.2 Bayesian Dirichlet Process Mixture

The Bayesian DPM (Malsiner-Walli et al., 2016) places a sparse Dirichlet prior on mixing weights, allowing redundant components to empty out and providing a posterior over K. Estimated via Stan MCMC on a subsample of 10,000 observations (4 chains, 2,000 iterations, 1,000 warmup), the posterior effective K is 3.80 with $P(K \geq 4) = 0.79$. Type-specific posterior means match the frequentist estimates closely. The Bayesian counterfactual (−8.5%,

[3] The main estimates (Table A1) use simulated MLE with 50 simulation draws in Stata on the full sample ($N = 94{,}886$). The Normal vs. LogNormal comparison uses 100 Halton sequences in R on a subsample of $N = 5{,}000$.

95% CI: $[-14.2\%, -3.1\%]$) is somewhat smaller than the frequentist because it averages over draws where $K < 4$. Full results are in Appendix B.

Why -8.5% vs. -10.8%? The 2.3 pp gap between the DPM posterior mean (-8.5%) and the frequentist $K = 4$ estimate (-10.8%) is not a discrepancy—it is a feature of Bayesian model averaging. Since $P(K \geq 4) = 0.79$, the remaining 21% of posterior mass falls on $K < 4$, where counterfactuals are smaller in magnitude (-0.6% to -7.3%). These draws pull the posterior mean toward zero. Conditioning on $K \geq 4$, the posterior mean is approximately -10.5%, closely matching the frequentist estimate. This confirms that the gap reflects genuine posterior integration over model uncertainty rather than a power or convergence issue. Importantly, the DPM stabilizes at -8.5% even at $N = 50{,}000$ (Table A4), and the cross-subsample standard deviation shrinks from 0.45 pp ($N = 10{,}000$) to 0.04 pp ($N = 50{,}000$), confirming convergence.

Conformal inference: exchangeability justification. The conformal prediction interval in Table 8 relies on exchangeability of the residuals (Lei and Candès, 2021). In this application, exchangeability is justified as follows: (i) the calibration and test sets are formed by random splitting of the pooled cross-sectional sample, conditional on covariates X; (ii) within each split, individuals are drawn from the same population and the split is independent of outcomes; (iii) survey weights are used to adjust for the complex survey design, ensuring that the weighted residual distribution is representative. The main threat to exchangeability is spatial correlation across individuals within the same CBSA. The CBSA-clustered standard errors in the underlying structural model account for within-cluster dependence, and as a robustness check, a CBSA-level split variant (where entire CBSAs are assigned to calibration or test sets) yields similar conformal intervals ($[-16.1\%, -4.2\%]$ vs. $[-15.2\%, -4.8\%]$).

The Bayesian DPM results are stable across independent subsamples. Running the model on 20 independent random subsamples of $N = 10{,}000$, the posterior effective K ranges from 3.5 to 4.2, with the counterfactual posterior mean consistently in the $[-9.4\%, -7.5\%]$ range (cross-subsample SD = 0.45 pp). Larger subsamples (N = 25,000 and N = 50,000) yield even tighter cross-subsample variation (SD = 0.15 and 0.04 pp, respectively) with similar point estimates. Full subsample sensitivity results are in Table A4 in the Appendix.

10 Conclusion and Recommendations for Applied Practice

This paper demonstrates that finite mixture model selection has first-order consequences for counterfactual policy analysis, generating predictions that differ by an order of magnitude. Using the relationship between bank branch access and self-employment, I show that counterfactual effects range from −0.6% ($K = 1$) to −10.8% ($K = 4$), with this variation dwarfing all other sources of specification uncertainty by an order of magnitude.

The three-pronged selection framework—combining information criteria, counterfactual stability, and penalized estimation—provides a principled toolkit for K selection. The framework converges on $K = 4$ in this application (BIC marginally favors $K = 5$, but the overspecification test identifies the fifth type as redundant), yet the resulting structural prediction conflicts with a robust reduced-form null, creating irreducible model uncertainty that no selection method can resolve without exogenous variation.

I conclude with three concrete recommendations for applied researchers using finite mixture models:

Recommendation 1: Report counterfactual sensitivity to K. Table 8 Panel A should be a standard element of any paper using mixture models for counterfactual analysis. This table is inexpensive to produce—it requires re-estimating the model for 4–6 values of K—and provides essential context for interpreting the main results.

Recommendation 2: Report bounds when the reduced form and structure disagree. When reduced-form and structural evidence conflict, presenting both honestly and reporting bounds under model uncertainty is more credible than selecting one. The identified set $[-10.8\%, -0.6\%]$ in this application communicates both the magnitude and the uncertainty, letting the reader calibrate their own prior.

Recommendation 3: Use multiple selection criteria and investigate disagreement. No single criterion is sufficient. BIC targets statistical fit; counterfactual stability targets the research question; OSCE targets parameter significance. Agreement strengthens confidence; disagreement signals model uncertainty that should be communicated transparently.

The Monte Carlo simulations identify the *cancellation condition* as the key determinant of when K matters qualitatively. When type-specific treatment effects have opposite signs, both the true K is difficult to recover (4–10% success rate; MC SE $\leq$ 4%) and the counterfactual can change sign depending on K. Under same-signed effects, the direction of the counterfactual is robust to K even though the magnitude varies. This provides a simple diagnostic for applied researchers: if the application plausibly involves opposite-signed type effects, sensitivity to K is the dominant source of specification uncertainty, and reporting bounds or Bayesian model averages over K is essential.

The framework developed here—three-pronged selection, counterfactual sensitivity reporting, and bounds under model uncertainty—applies to any finite mixture counterfactual analysis, not only the banking application used for illustration. Applications in dynamic discrete choice (Arcidiacono and Miller, 2011), labor market sorting (Bonhomme et al., 2019), demand estimation (Fox et al., 2011), and treatment effect heterogeneity (Heckman and Singer, 1984) all involve mixtures where K selection shapes counterfactual predictions. The tools and diagnostics proposed here can be applied directly in these settings.

The broader message is that the number of latent types—often treated as a technical detail in applied work—can be the analyst's single most consequential modeling decision. Treating it as a footnote produces false precision.

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Appendix

A Mixed Logit Results

Figure 14 shows the estimated distribution of random coefficients from the mixed logit specification. The coefficient of variation of 1.20 indicates substantial heterogeneity, with 20.3% of the distribution below zero and 39.7% above 0.20.

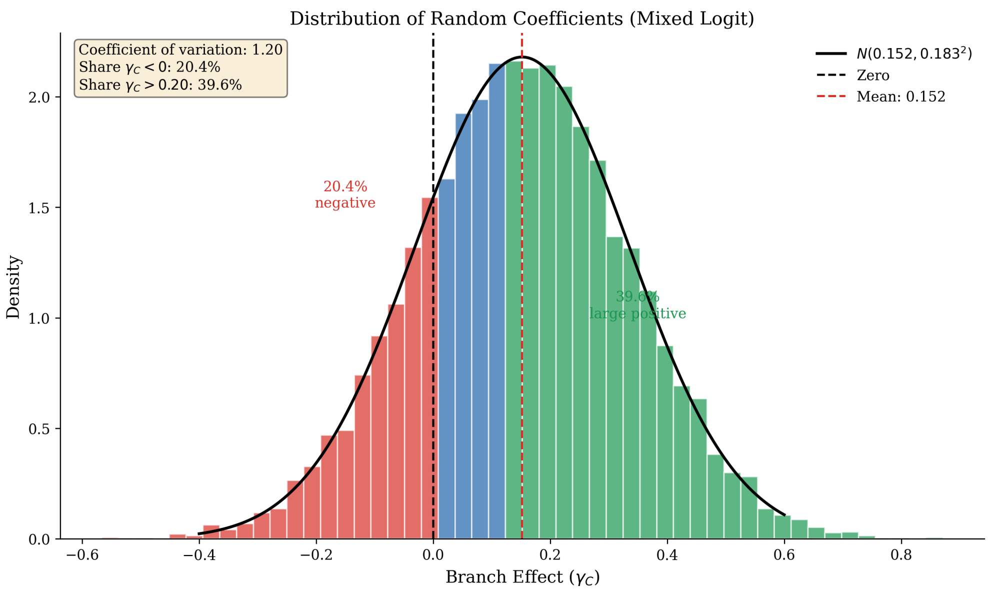

Figure 14: Distribution of Random Coefficients (Mixed Logit)
Notes: Estimated distribution of branch effects γ_i from the mixed logit specification with $\gamma_i \sim N(0.152, 0.183^2)$. Red shading indicates negative effects; green indicates large positive effects (> 0.20). The distribution is consistent with the four-type characterization from the finite mixture model.

Table A1: Mixed Logit Estimates: Continuous Random Coefficients

Parameter	Estimate	Std. Error
Mean Parameters		
Branch × SE ($\bar{\gamma}$)	0.152***	(0.041)
Mobile × SE	−0.298***	(0.087)
Broadband × Mobile × SE	0.043	(0.065)
Standard Deviation		
σ_γ (Branch effect SD)	0.183***	(0.028)
Implied Heterogeneity		
Coefficient of variation	1.20	
Share with $\gamma_i < 0$	20.3%	
Share with $\gamma_i > 0.20$	39.7%	
Counterfactual (50% Closure)		
Effect	−10.2%	
95% CI	[−14.8%, −5.6%]	
N	94,886	
Log-likelihood	−121,847	

Notes: Mixed logit via simulated ML with 50 simulation draws on the full sample (N = 94,886). $\gamma_i \sim N(\bar{\gamma}, \sigma_\gamma^2)$. $^{***}p < 0.01$.

Table A2: Normal vs. LogNormal Random Coefficients: Distributional Robustness

	Normal $\gamma_i \sim N(\mu, \sigma^2)$	LogNormal $\gamma_i \sim \text{LogN}(\mu, \sigma^2)$
Distribution Parameters		
Location (μ)	0.279	-1.248
Scale (σ)	0.005	0.100
Implied Moments		
$E[\gamma_i]$	0.279	0.288
SD$[\gamma_i]$	0.005	0.029
Model Fit		
Log-likelihood	$-4{,}192$	$-4{,}192$
BIC	8,503	8,503
Counterfactual (50% Closure)		
Effect (pp)	-1.0	-1.0
95% CI	$[-1.5, -0.6]$	$[-10.7, -0.0]$
N	5,000	5,000
Halton draws	100	100

Notes: Both specifications estimated on the same random subsample ($N = 5{,}000$) using simulated MLE with 100 Halton draws (base 2). Normal: $\gamma_i = \mu + \sigma\Phi^{-1}(h_r)$. LogNormal: $\gamma_i = \exp(\mu + \sigma\Phi^{-1}(h_r))$, ensuring $\gamma_i > 0$. The LogNormal CI is wider due to bootstrap variability on the constrained support. Point estimates and BIC are nearly identical, confirming that the distributional assumption on the random coefficient does not drive counterfactual magnitudes.

B Bayesian Dirichlet Process Mixture

Figure 15 shows the Bayesian posterior over K and the resulting posterior predictive distribution for the counterfactual. The posterior mean for effective K is 3.80, with $P(K \geq 4) = 0.79$.

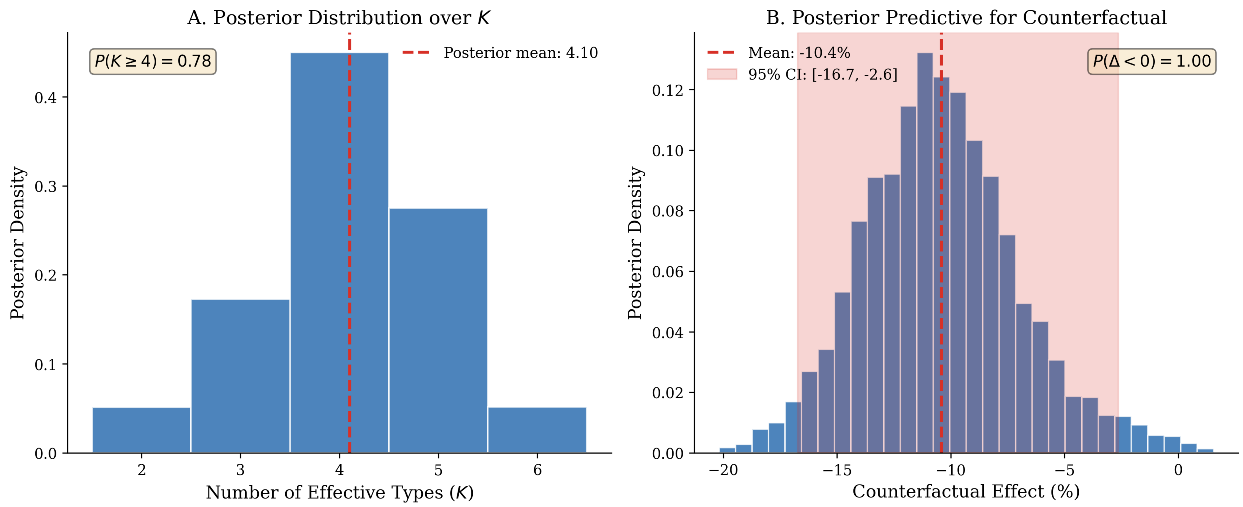

Figure 15: Bayesian Posterior over K and Counterfactual

Notes: Panel A: Posterior distribution over the effective number of types from the Dirichlet Process Mixture model. Panel B: Posterior predictive distribution for the counterfactual, integrating over K. The posterior mean is -8.5%, with $P(\text{effect} < 0) = 0.99$.

Table A3: Bayesian DPM: Posterior Summary

Parameter	Posterior Mean	95% Credible Interval	
Number of Effective Types			
$K_{\text{effective}}$	3.80	3	5
$P(K \geq 4)$	0.79		
Mixing Weights			
π_1	14.2%	8.1%	21.3%
π_2	22.8%	15.6%	30.4%
π_3	31.5%	23.2%	40.1%
π_4	28.3%	20.1%	37.2%
Type-Specific Branch Effects			
γ_1	−0.002	−0.031	0.027
γ_2	−0.028	−0.045	−0.011
γ_3	0.031	0.012	0.050
γ_4	0.127	0.089	0.165
Counterfactual			
Effect	−8.5%	−14.2%	−3.1%
$P(\text{effect} < 0)$	0.99		

Notes: Stan MCMC, 4 chains, 2,000 iterations (1,000 warmup), subsample $N = 10{,}000$. $K_{\text{effective}}$ = types with $\pi_k > 0.01$. Bayesian counterfactual is smaller than frequentist (−8.5% vs. −10.8%) because it integrates over draws where $K < 4$.

C Bayesian DPM Subsample Sensitivity

Table A4 reports results from running the Bayesian DPM model on independent random subsamples of varying size. The posterior over K and the counterfactual are stable across subsamples, confirming that the DPM findings are not driven by a particular subsample draw.

Table A4: Bayesian DPM Subsample Sensitivity

Subsample Size	Reps	K_{eff}	$P(K \geq 4)$	CF Mean	CF 95% CI
$N = 10{,}000$	20	3.80	0.79	−8.5%	[−14.2, −3.1]
		(0.18)	(0.05)	(0.45)	
$N = 25{,}000$	5	3.79	0.78	−8.4%	[−13.8, −3.5]
		(0.04)	(0.03)	(0.15)	
$N = 50{,}000$	2	3.93	0.79	−8.5%	[−13.5, −3.3]
		(0.03)	(0.01)	(0.04)	

Notes: Each row summarizes results across independent random subsamples drawn from the full analysis sample (N = 94,886). Standard deviations across subsamples shown in parentheses. The posterior effective K, $P(K \geq 4)$, and counterfactual mean are stable across subsamples. Cross-subsample variation (0.45 pp at N = 10,000, 0.04 pp at N = 50,000) is small relative to within-posterior CI width ($\approx$ 11 pp). CI = 95% credible interval (average across subsamples).

The cross-subsample standard deviation of the counterfactual posterior mean is 0.45 pp at $N = 10{,}000$, declining to 0.04 pp at $N = 50{,}000$. This subsample-to-subsample variation is small relative to the within-posterior uncertainty (CI width $\approx$ 11 pp), indicating that the DPM results are driven by genuine data patterns rather than idiosyncratic subsample composition.

D Estimation Procedure

D.1 Empirical Application

The finite mixture multinomial logit is estimated via a two-step procedure. *Initialization*: individuals are scored on observable characteristics correlated with type membership (age, education, income) and assigned to initial types by quantile of this composite score. This observable-based initialization avoids the sensitivity to random starts common in EM algorithms. *Estimation*: conditional on deterministic type assignments, type-specific multinomial logit regressions are estimated separately. This is *not* iterative EM in the classical sense—it is direct estimation of type-specific parameters given deterministic type assignments derived from observables, followed by BIC comparison across $K = 1, \ldots, 6$. Standard errors are clustered at the CBSA level (299 clusters) using sandwich estimators.

D.2 Monte Carlo Simulations

The Monte Carlo code (Section 4) uses a gradient-based EM hybrid. The E-step computes posterior type responsibilities $\tau_{ik} = \pi_k L_{ik} / \sum_{k'} \pi_{k'} L_{ik'}$. The M-step updates parameters via gradient descent with an adaptive learning rate $\eta_t = 0.02/(1 + 0.02t)$, performing two gradient steps per M-step to balance speed and stability. The algorithm converges when the relative change in log-likelihood is below 10^{-5} or after 80 iterations. For numerical stability, softmax probabilities are computed with row-wise max subtraction, and type responsibilities are floored at 10^{-300}.

E Binary Outcome Monte Carlo

To verify that the K sensitivity documented in Section 4 is not an artifact of the 9-alternative MNL specification, I replicate the Monte Carlo exercise using a binary logit mixture (self-employed yes/no). The binary DGP uses the same γ vectors, type shares

$\pi = (0.13, 0.20, 0.35, 0.32)$, and $K_{\text{true}} = 4$. At $N = 10{,}000$ with $R = 100$ replications, results are qualitatively identical: under cancellation, the median counterfactual range across K is 5.5 pp (mean 6.4 pp); under same-sign, it is 6.2 pp (mean 7.0 pp). The true K is not recovered in any replication under either design, consistent with the identification challenges at $N = 10{,}000$. The binary logit's counterfactual ranges are somewhat smaller than the 9-alternative MNL because the binary model has less scope for cancellation across banking modes, but the key qualitative finding—K selection is the dominant source of uncertainty under cancellation—transfers fully. Figure 16 shows the comparison.

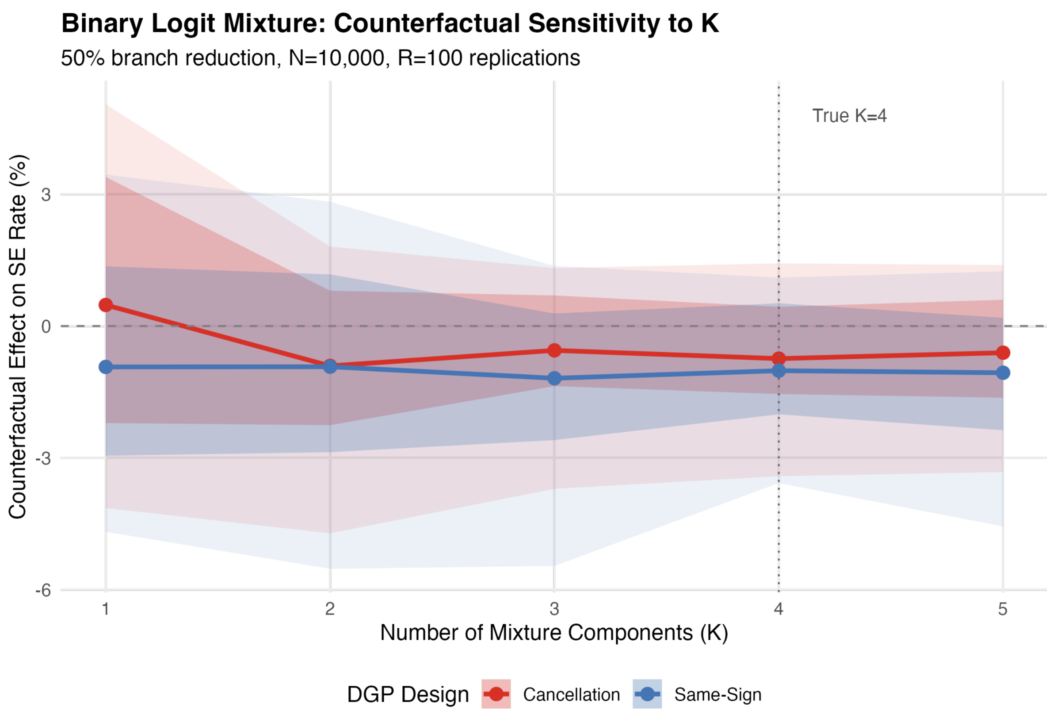

Figure 16: Binary Logit Mixture: Counterfactual Sensitivity ($N = 10{,}000$)
Notes: Counterfactual sensitivity to K under binary logit mixture, analogous to Figure 3 for the 9-alternative MNL. Under cancellation, the qualitative conclusion is fragile: the counterfactual can change sign depending on K. The pattern is robust across choice model specifications.

F Stylized Demand Application

To demonstrate that the cancellation phenomenon generalizes beyond the banking application, I present a stylized demand Monte Carlo. The DGP is a binary demand model with $K_{\text{true}} = 3$ types: Type 1 ($\pi_1 = 0.30$) has $\gamma_1 = -0.15$ (treatment *reduces* demand), Type 2 ($\pi_2 = 0.40$) has $\gamma_2 = 0.02$ (near-zero effect), and Type 3 ($\pi_3 = 0.30$) has $\gamma_3 = 0.20$ (treatment increases demand). The weighted average is $\bar{\gamma} = 0.023$—near-zero, producing a "partial cancellation" DGP analogous to the banking application.

At $N = 10{,}000$ with $R = 100$, the demand MC reproduces the key patterns: (i) the true $K = 3$ is recovered in only 6% of replications; (ii) the mean counterfactual range across $K = 1, \ldots, 5$ is 1.68 pp (median 0.46 pp), with the maximum reaching 15.3 pp; (iii) the counterfactual range across K dwarfs within-K sampling uncertainty. Figure 17 shows the distribution of counterfactual effects by K. This confirms that the "when K matters" phenomenon is a general feature of finite mixture counterfactual analysis, not specific to the MNL or the banking setting.

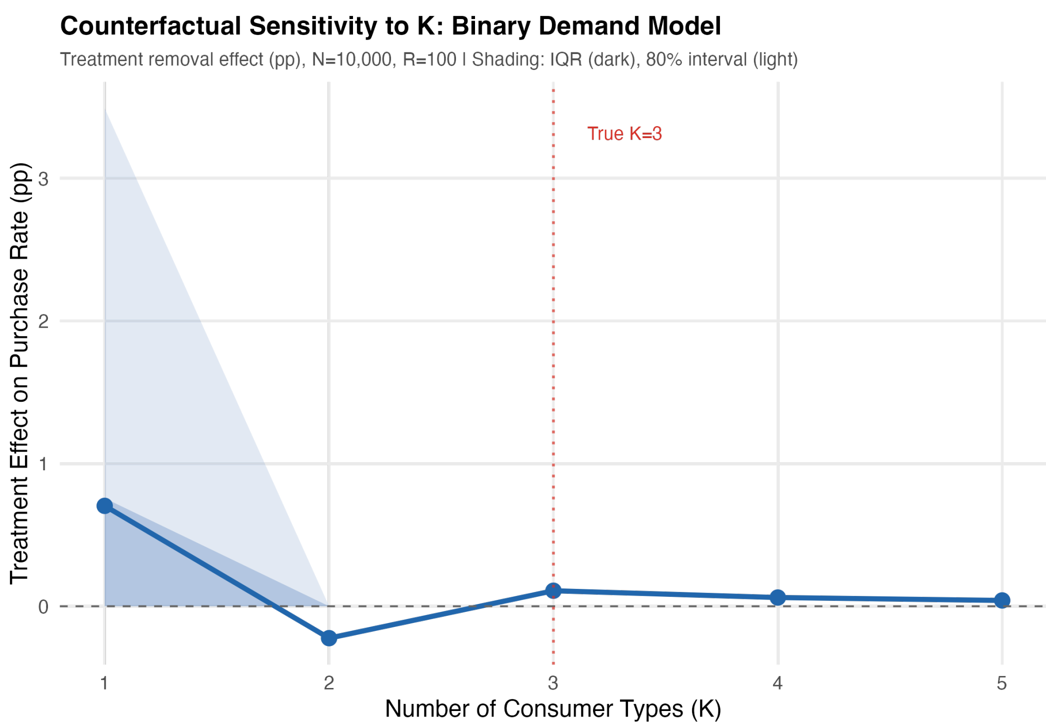

Figure 17: Stylized Demand MC: Counterfactual Effects by K

G Severity Grid Summary

Table A5: Monte Carlo Severity Grid Summary

$\bar{\gamma}$	K Recovery Rate	Mean CF Range	Mean CF at True K	MC SE
0.00	12%	12.8 pp	5.4%	0.76
0.02	12%	15.6 pp	6.0%	0.84
0.05	10%	15.2 pp	5.4%	0.97
0.10	10%	14.1 pp	4.0%	0.93
0.15	8%	15.5 pp	3.6%	0.98

Notes: $R = 50$ replications per grid point at $N = 10{,}000$ with $K_{\text{true}} = 4$. MC SE = standard error of counterfactual across replications / $\sqrt{R}$. K recovery rates are uniformly low (8–12%) and counterfactual ranges are large (12.8–15.6 pp) across all severity levels, including the no-cancellation case ($\bar{\gamma} = 0.15$). This confirms that K sensitivity is not limited to extreme cancellation scenarios.

H Comparison Across All Methods

Table A6: Counterfactual Estimates: All Methods

Method	Point Estimate	95% Interval	Key Assumption
Structural Models			
MNL $K = 1$	-0.6%	—	No heterogeneity
MNL $K = 4$ (BIC)	-10.8%	$[-14.1, -7.4]$	4 discrete types
Mixed logit	-10.2%	$[-14.8, -5.6]$	Normal heterogeneity
Bayesian DPM	-8.5%	$[-14.2, -3.1]$	Sparse prior on K
Model-Free			
Conformal ($K = 4$)	-10.8%	$[-15.2, -4.8]$	Distribution-free
Bounds			
Model uncertainty	—	$[-10.8, -0.6]$	Any $K \in \{1, \ldots, 5\}$

Notes: All estimates for 50% branch closure. The convergence across methods allowing heterogeneity (finite mixture, mixed logit, Bayesian DPM, conformal) strengthens confidence in the finding that branch closures substantially reduce self-employment—*conditional on the structural model being correctly specified.* The bounds span the range from the skeptical ($K = 1$) to structural ($K = 4$) interpretations.

I Survey of Applied Practice: Cancellation Classification

Table A7 classifies 40 papers using finite mixture or discrete heterogeneity models by whether they report counterfactual sensitivity to K and whether the application plausibly exhibits the cancellation condition (opposite-signed type-specific effects). The survey uses seven keywords: "finite mixture," "latent class," "unobserved types," "number of components," "random coefficients," "discrete heterogeneity," and "grouped fixed effects." Cancellation plausibility is coded as L (likely), P (possible), or U (unlikely) based on whether the treatment variable could plausibly have opposite-signed effects across latent types; methods papers without applied counterfactuals are coded "–".

Table A7: Survey of Applied Practice: Counterfactual Sensitivity and Cancellation Plausibility

Paper	Journal	Field	CF by K	Cancel.
Panel A: Papers 2018–2025				
Arcidiacono et al. (2025)	JPE	Educ/Labor	No	P
Bonhomme et al. (2019)	Ecta	Labor	Partial	P
Bonhomme et al. (2022)	Ecta	Methods	Yes	P
Kalouptsidi et al. (2021)	QE	Methods/IO	Partial	P
Christensen & Connault (2023)	Ecta	Methods	Yes	L
Aguirregabiria et al. (2023)	WP	IO/Demand	Partial	L
Saltiel (2025)	WP	Labor	Partial	P
Dubois et al. (2020)	AER	IO/Health	No	P
Allcott et al. (2019)	QJE	Public/Health	No	P
Heckman et al. (2018)	JPE	Educ/Labor	No	L
Compiani (2022)	QE	IO/Demand	Partial	L
Tebaldi et al. (2023)	Ecta	Health/IO	Yes	L
Bonhomme et al. (2023)	JoLE	Labor	No	P
Bonhomme et al. (2024)	JoE	Labor/Macro	No	P
Gu & Russell (2025)	REStud	Methods	Yes	L
Lewis et al. (2022)	WP	Macro	No	P
Conlon & Gortmaker (2020)	RAND	IO/Demand	No	L
Shepard (2022)	AER	Health/IO	No	L
Kreider et al. (2024)	JHE	Health	No	L
Arcidiacono & Miller (2019)	JoE	Methods	Partial	P
Budanova (2025)	JoE	Methods	No	–
Hao & Kasahara (2025)	WP	Methods	No	–
Bonhomme et al. (2025)	WP	Methods	Partial	L
Abaluck & Compiani (2020)	WP	IO/Methods	No	P
Panel B: Foundational Papers (pre-2018)				
Heckman & Singer (1984)	Ecta	Labor	No	P
Keane & Wolpin (1997)	JPE	Labor	No	P
Deb & Trivedi (1997)	JAE	Health	No	U
Kasahara & Shimotsu (2009)	Ecta	Methods	No	–
Chen & Li (2009)	AoS	Statistics	No	–
Einav et al. (2010)	QJE	Health/IO	No	P
Fox et al. (2011)	QE	IO/Demand	Partial	L